





## **The Project Gutenberg eBook of Sixth annual report of the Bureau of ethnology. (1888 N 06 / 1884-1885)**

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at [www.gutenberg.org](http://www.gutenberg.org). If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Sixth annual report of the Bureau of ethnology. (1888 N 06 / 1884-1885)

Editor: John Wesley Powell

Contributor: Franz Boas

James Owen Dorsey

William Henry Holmes

Cyrus Thomas

Release date: March 8, 2016 [eBook #51390]

Most recently updated: October 23, 2024

Language: English

Other information and formats: [www.gutenberg.org/ebooks/51390](http://www.gutenberg.org/ebooks/51390)

Credits: Richard Tonsing, PM for Bureau of American Ethnology and the  
Online Distributed Proofreading Team

\*\*\* START OF THE PROJECT GUTENBERG EBOOK SIXTH ANNUAL  
REPORT OF THE BUREAU OF ETHNOLOGY. (1888 N 06 / 1884-1885)

\*\*\*

# LIBRARY CATALOGUE SLIPS.

Series Title.

**Smithsonian institution.** *Bureau of ethnology.*

Sixth annual report | of the | Bureau of ethnology | to the |  
secretary of the Smithsonian institution | 1884-'85 | by | J. W.  
Powell | director | [Vignette] |

Washington | government printing office | 1888

8°. lviii, 675 pp. incl. 15 pl. & 4 pp. of music. 8 pl. 2 maps.

Author title.

**Powell** (John Wesley).

Sixth annual report | of the | Bureau of ethnology | to the |  
secretary of the Smithsonian institution | 1884-'85 | by | J. W.  
Powell | director | [Vignette] |

Washington | government printing office | 1888

8°. lviii, 675 pp. incl. 15 pl. & 4 pp. of music. 8 pl. 2 maps.

[SMITHSONIAN INSTITUTION. *Bureau of ethnology.*]

Title for  
subject  
entry.

Sixth annual report | of the | Bureau of ethnology | to the |  
secretary of the Smithsonian institution | 1884-'85 | by | J. W.  
Powell | director | [Vignette] |

Washington | government printing office | 1888

8°. lviii, 675 pp. incl. 15 pl. & 4 pp. of music. 8 pl. 2 maps.

[SMITHSONIAN INSTITUTION. *Bureau of ethnology.*]

**SIXTH ANNUAL REPORT**  
**OF THE**  
**BUREAU OF ETHNOLOGY**  
**TO THE**  
**SECRETARY OF THE SMITHSONIAN INSTITUTION**  
**1884-'85**

**BY**  
**J.W. POWELL**  
**DIRECTOR**



WASHINGTON  
GOVERNMENT PRINTING OFFICE  
1888

# CONTENTS.

## REPORT OF THE DIRECTOR.

|                                 | <b>Page.</b>   |
|---------------------------------|----------------|
| Letter of transmittal           | <u>XXI</u>     |
| Introduction                    | <u>XXIII</u>   |
| Publication                     | <u>XXIV</u>    |
| Field work                      | <u>XXVI</u>    |
| Mound explorations              | <u>XXVI</u>    |
| Work of Prof. Cyrus Thomas      | <u>XXVI</u>    |
| Explorations in the Southwest   | <u>XXVIII</u>  |
| Work of Mr. James Stevenson     | <u>XXVIII</u>  |
| Work of Mr. Victor Mindeleff    | <u>XXIX</u>    |
| Linguistic field work           | <u>XXX</u>     |
| Work of Mrs. Erminnie A. Smith  | <u>XXX</u>     |
| Work of Mr. H. W. Henshaw       | <u>XXXI</u>    |
| Work of Mr. A. S. Gatschet      | <u>XXXIII</u>  |
| Work of Rev. J. Owen Dorsey     | <u>XXXVI</u>   |
| Work of Mr. Jeremiah Curtin     | <u>XXXVII</u>  |
| General field work              | <u>XXXVIII</u> |
| Work of Dr. Washington Matthews | <u>XXXVIII</u> |
| Work of Dr. H. C. Yarrow        | <u>XL</u>      |

|                                                                                                             |                               |
|-------------------------------------------------------------------------------------------------------------|-------------------------------|
| Work of Dr. W. J. Hoffman                                                                                   | <a href="#"><u>XL</u></a>     |
| Office work                                                                                                 | <a href="#"><u>XLIII</u></a>  |
| Work of Mr. H. W. Henshaw                                                                                   | <a href="#"><u>XLV</u></a>    |
| Work of Mrs. Erminnie A. Smith                                                                              | <a href="#"><u>XLV</u></a>    |
| Work of Col. Garrick Mallery                                                                                | <a href="#"><u>XLV</u></a>    |
| Work of Dr. W. J. Hoffman                                                                                   | <a href="#"><u>XLV</u></a>    |
| Work of Mr. James C. Pilling                                                                                | <a href="#"><u>XLV</u></a>    |
| Work of Mr. Frank H. Cushing                                                                                | <a href="#"><u>XLVI</u></a>   |
| Work of Prof. Cyrus Thomas                                                                                  | <a href="#"><u>XLVII</u></a>  |
| Work of Mr. Victor Mindeleff                                                                                | <a href="#"><u>XLVII</u></a>  |
| Work of Rev. J. Owen Dorsey                                                                                 | <a href="#"><u>XLVIII</u></a> |
| Work of Mr. Albert S. Gatschet                                                                              | <a href="#"><u>XLVIII</u></a> |
| Work of Mr. W. H. Holmes                                                                                    | <a href="#"><u>XLVIII</u></a> |
| Work of Dr. H. C. Yarrow                                                                                    | <a href="#"><u>L</u></a>      |
| Work of Mr. Charles C. Royce                                                                                | <a href="#"><u>L</u></a>      |
| Accompanying papers                                                                                         | <a href="#"><u>LI</u></a>     |
| Ancient art of the province of Chiriqui, Colombia, by<br>William H. Holmes                                  | <a href="#"><u>LI</u></a>     |
| A study of the textile art in its relation to the development of<br>form and ornament, by William H. Holmes | <a href="#"><u>LIV</u></a>    |
| Aids to the study of the Maya Codices, by Cyrus Thomas                                                      | <a href="#"><u>LV</u></a>     |
| Osage traditions, by Rev. J. Owen Dorsey                                                                    | <a href="#"><u>LVI</u></a>    |
| The Central Eskimo, by Dr. Franz Boas                                                                       | <a href="#"><u>LVI</u></a>    |
| Financial statement                                                                                         | <a href="#"><u>LVIII</u></a>  |

## ACCOMPANYING PAPERS.

### ANCIENT ART OF THE PROVINCE OF CHIRIQUI, BY WILLIAM H. HOLMES.

|                    | <b>Page.</b> |
|--------------------|--------------|
| Introduction       | 13           |
| Geography          | 13           |
| Literature         | 14           |
| People             | 15           |
| The cemeteries     | 16           |
| The graves         | 17           |
| Human remains      | 20           |
| Placing of relics  | 21           |
| <br>Objects of art | <br>21       |
| Stone              | 21           |
| Pictured rocks     | 21           |
| Columns            | 22           |
| Images             | 23           |
| Mealing stones     | 25           |
| Stools             | 27           |
| Celts etc.         | 29           |
| Spearheads         | 34           |
| Arrowpoints        | 34           |
| Ornaments          | 34           |
| Metal              | 35           |
| Gold and copper    | 35           |
| Bronze             | 49           |
| Clay: Pottery      | 53           |

|                             |     |
|-----------------------------|-----|
| Preliminary                 | 53  |
| How found                   | 55  |
| Material                    | 55  |
| Manufacture                 | 56  |
| Color                       | 57  |
| Use                         | 57  |
| Forms of vessels            | 58  |
| Decoration                  | 62  |
| Unpainted ware              | 66  |
| Terra cotta group           | 67  |
| Black incised group         | 80  |
| Painted ware                | 84  |
| Scarified group             | 87  |
| Handled group               | 90  |
| Tripod group                | 97  |
| Maroon group                | 107 |
| Red line group              | 109 |
| White line group            | 111 |
| Lost color group            | 113 |
| Alligator group             | 130 |
| Polychrome group            | 140 |
| Unclassified                | 147 |
| Miscellaneous objects       | 149 |
| Spindle whorls              | 149 |
| Needlecases                 | 150 |
| Figurines                   | 151 |
| Stools                      | 154 |
| Musical instruments         | 156 |
| Rattles                     | 156 |
| Drums                       | 157 |
| Wind instruments            | 160 |
| Life forms in vase painting | 171 |

**A STUDY OF THE TEXTILE ART IN ITS RELATION TO THE  
DEVELOPMENT OF FORM AND ORNAMENT, BY WILLIAM  
H. HOLMES.**

|                                                      |     |
|------------------------------------------------------|-----|
| Introduction                                         | 195 |
| Form in textile art                                  | 196 |
| Relations of form to ornament                        | 201 |
| Color in textile art                                 | 201 |
| Textile ornament                                     | 202 |
| Development of a geometric system within the art     | 202 |
| Introduction                                         | 202 |
| Relief phenomena                                     | 203 |
| Ordinary features                                    | 203 |
| Reticulated work                                     | 210 |
| Superconstructive features                           | 211 |
| Color phenomena                                      | 215 |
| Ordinary features                                    | 215 |
| Non-essential constructive features                  | 226 |
| Superconstructive features                           | 228 |
| Adventitious features                                | 231 |
| Geometricity imposed upon adopted elements of design | 232 |
| Extension of textile ornament to other forms of art  | 244 |

**AIDS TO THE STUDY OF THE MAYA CODICES, BY CYRUS  
THOMAS.**

|                                            |     |
|--------------------------------------------|-----|
| Introduction                               | 259 |
| CHAP. I. The numerals in the Dresden Codex | 261 |
| II. Conclusions                            | 339 |
| III. The writing                           | 345 |

|                                                      |     |
|------------------------------------------------------|-----|
| Signification of the characters                      | 347 |
| Symbols of animals etc                               | 348 |
| Symbols of deities                                   | 358 |
| Discussion as to phonetic features of the characters | 365 |

## **OSAGE TRADITIONS, BY REV. J. OWEN DORSEY.**

|                                                                                   |     |
|-----------------------------------------------------------------------------------|-----|
| Introduction                                                                      | 377 |
| Traditions of the elders                                                          | 381 |
| Unŭ <sup>n</sup> uꞑáꞑe. Tsícu wactáꞑe itáde (Tradition of the Tsícu wactáꞑe gens) | 381 |
| Translation                                                                       | 388 |
| Unŭ <sup>n</sup> uꞑáꞑe. Qũꞑápasan itáde (Tradition of the Bald Eagle subgens)     | 390 |
| Translation                                                                       | 394 |
| Concluding remarks                                                                | 396 |

## **THE CENTRAL ESKIMO, BY DR. FRANZ BOAS.**

|                                   |     |
|-----------------------------------|-----|
| Introduction                      | 409 |
| Authorities quoted                | 410 |
| Orthography                       | 413 |
| Geography of northeastern America | 413 |
| Distribution of the tribes        | 419 |
| General observations              | 419 |
| Baffin Land                       | 421 |
| The Sikosuilarmiut                | 421 |
| The Akuliarmiut                   | 421 |
| The Qaumauangmiut                 | 421 |
| The Nugumiut                      | 422 |
| The Oqomiut                       | 424 |

|                                                                                   |         |
|-----------------------------------------------------------------------------------|---------|
| The Padlimiut and the Akudnirmiut                                                 | 440     |
| The Aggomiut                                                                      | 442     |
| The Iglulirmiut                                                                   | 444     |
| The Pilingmiut                                                                    | 444     |
| The Sagdlirmiut                                                                   | 444     |
| Western shore of Hudson Bay                                                       | 444     |
| The Aivillirmiut                                                                  | 445     |
| The Kinipetu or Agutit                                                            | 450     |
| The Sagdlirmiut of Southampton Island                                             | 451     |
| The Sinimiut                                                                      | 451     |
| Boothia Felix and Back River                                                      | 452     |
| The Netchillirmiut                                                                | 452     |
| The Ugjulirmiut                                                                   | 458     |
| The Ukusiksalirmiut                                                               | 458     |
| Smith Sound                                                                       | 459     |
| The natives of Ellesmere Land                                                     | 459     |
| The North Greenlanders                                                            | 460     |
| <br>Influence of geographical conditions upon the distribution of the settlements | <br>460 |
| <br>Trade and intercourse between the tribes                                      | <br>462 |
| <br>List of the Central Eskimo tribes                                             | <br>470 |
| <br>Hunting and fishing                                                           | <br>471 |
| Seal, walrus, and whale hunting                                                   | 471     |
| Deer, musk ox, and bear hunting                                                   | 501     |
| Hunting of small game                                                             | 510     |
| Fishing                                                                           | 513     |
| <br>Manufactures                                                                  | <br>516 |

|                                                               |     |
|---------------------------------------------------------------|-----|
| Making leather and preparing skins                            | 516 |
| Sundry implements                                             | 523 |
| Transportation by boats and sledges                           | 527 |
| The boat                                                      | 527 |
| The sledge and dogs                                           | 529 |
| Habitations and dress                                         | 539 |
| The house                                                     | 539 |
| Clothing, dressing of the hair, and tattooing                 | 554 |
| Social and religious life                                     | 561 |
| Domestic occupations and amusements                           | 561 |
| Visiting                                                      | 574 |
| Social customs in summer                                      | 576 |
| Social order and laws                                         | 578 |
| Religious ideas and the angakunirn (priesthood)               | 583 |
| Sedna and the fulmar                                          | 583 |
| The tornait and the angakut                                   | 591 |
| The flight to the moon                                        | 598 |
| Kadlu the thunderer                                           | 600 |
| Feasts, religious and secular                                 | 600 |
| Customs and regulations concerning birth, sickness, and death | 609 |
| Tales and traditions                                          | 615 |
| Ititaujang                                                    | 615 |
| The emigration of the Sagdlirmiut                             | 618 |
| Kalopaling                                                    | 20  |
| The Uissuit                                                   | 621 |
| Kiviung                                                       | 621 |
| The origin of the narwhal                                     | 625 |

|                                                                     |     |
|---------------------------------------------------------------------|-----|
| The visitor                                                         | 627 |
| The fugitive women                                                  | 628 |
| Qaudjaqdjuq                                                         | 628 |
| I. Story of the three brothers                                      | 628 |
| II. Qaudjaqdjuq                                                     | 630 |
| Igimarasugdjuqdjuq the cannibal                                     | 633 |
| The Tornit                                                          | 634 |
| The woman and the spirit of the singing house                       | 636 |
| The constellation Udleqdjun                                         | 636 |
| The origin of the Adlet and of the Qadluait                         | 637 |
| The great flood                                                     | 637 |
| Inugpaqdjuqdjualung                                                 | 638 |
| The bear story                                                      | 638 |
| Sundry tales                                                        | 639 |
| Tables relating to animals                                          | 641 |
| The owl and the raven                                               | 641 |
| Comparison between Baffin Land traditions and those of other tribes | 641 |
| Science and the arts                                                | 643 |
| Geography and navigation                                            | 643 |
| Poetry and music                                                    | 648 |
| Merry-making among the Tornit                                       | 649 |
| The lemming's song                                                  | 650 |
| Arlum pissinga (the killer's song)                                  | 650 |
| I. Summer song                                                      | 653 |
| II. The returning hunter                                            | 653 |
| III. Song of the Tornit                                             | 653 |
| IV. Song of the Inuit traveling to Nettiilling                      | 653 |
| V. Oxaitoq's song                                                   | 654 |
| VI. Utitiaq's song                                                  | 654 |
| VII. Song                                                           | 654 |

|                          |             |
|--------------------------|-------------|
| VIII. Song               | 654         |
| IX. Song of the Tornit   | 654         |
| X. The fox and the woman | 655         |
| XI. The raven's song     | 655         |
| XII. Song of a Padlimio  | 655         |
| XIII. Ititaujang's song  | 655         |
| XIV. Playing at ball     | 656         |
| XV. Playing at ball      | 657         |
| XVI.-XIX. Extracts       | 657,<br>658 |
| Glossary                 | 659         |
| Appendix                 | 667         |

## INDEX.

|       |                     |
|-------|---------------------|
| Index | <a href="#">671</a> |
|-------|---------------------|

## ILLUSTRATIONS.

|                                                                                                                      | Page.               |
|----------------------------------------------------------------------------------------------------------------------|---------------------|
| PLATE I. Map of Chiriqui                                                                                             | 13                  |
| II. Map showing in detail the geographical divisions of territory occupied by the Eskimo tribes of northeast America | <a href="#">[1]</a> |
| 1. Oqo and Akudnirn.                                                                                                 |                     |
| 2. Frobisher Bay.                                                                                                    |                     |
| 3. Eclipse Sound and Admiralty Inlet.                                                                                |                     |
| 4. Repulse Sound and Lyon Inlet.                                                                                     |                     |
| 5. Boothia Isthmus and King William Land.                                                                            |                     |
| III. Map of the territory occupied by the Eskimo tribes of North America, showing the boundaries                     | <a href="#">[1]</a> |
| IV. Cumberland Peninsula, drawn by Aranin, a Saumingmio                                                              | 643                 |
| V. Eskimo drawings                                                                                                   | 648                 |
| VI. Eskimo drawings                                                                                                  | 650                 |
| VII. Eskimo drawings                                                                                                 | 651                 |

|         |                                                                 |     |
|---------|-----------------------------------------------------------------|-----|
| VIII.   | Eskimo carvings                                                 | 652 |
| IX.     | Eskimo carvings                                                 | 653 |
| X.      | Modern Eskimo implements                                        | 654 |
| FIG. 1. | Section of oval grave                                           | 17  |
| 2.      | Section of a quadrangular grave                                 | 18  |
| 3.      | Grave with pillars                                              | 18  |
| 4.      | Compound cist                                                   | 19  |
| 5.      | Southwest face of the pictured stone                            | 22  |
| 6.      | A goddess of the ancient Chiriquians                            | 23  |
| 7.      | A god of the ancient Chiriquians                                | 24  |
| 8.      | Fragmentary human figure in gray basaltic rock                  | 25  |
| 9.      | Mealing stone with large tablet ornamented with<br>animal heads | 26  |
| 10.     | Puma shaped metate                                              | 27  |
| 11.     | Stool shaped object                                             | 28  |
| 12.     | Stool with columnar base                                        | 28  |
| 13.     | Stool with perforated base                                      | 29  |
| 14.     | Large partially polished celt                                   | 30  |
| 15.     | Celt of hexagonal section                                       | 31  |
| 16.     | Small wide bladed celt                                          | 31  |
| 17.     | Celt with heavy shaft                                           | 31  |
| 18.     | Celt or ax with constriction near the top                       | 31  |
| 19.     | Flaked and partially polished celt                              | 32  |
| 20.     | Well polished celt                                              | 32  |
| 21.     | Narrow pointed celt                                             | 32  |
| 22.     | Narrow pointed celt                                             | 32  |
| 23.     | Cylindrical celt with narrow point                              | 33  |

|     |                                                       |    |
|-----|-------------------------------------------------------|----|
| 24. | Leaf shaped objects suggesting spearpoints            | 34 |
| 25. | Arrowpoints                                           | 34 |
| 26. | Human figure formed of copper-gold alloy              | 41 |
| 27. | Grotesque human figure in gold                        | 42 |
| 28. | Rudely shaped human figure in gold                    | 42 |
| 29. | Grotesque human figure in nearly pure copper          | 43 |
| 30. | Grotesque human figure in nearly pure gold            | 43 |
| 31. | Rudely executed image of a bird in gold               | 44 |
| 32. | Image of a bird in gold                               | 45 |
| 33. | Puma shaped figure in gold                            | 45 |
| 34. | Puma shaped figure in base metal                      | 45 |
| 35. | Quadruped with grotesque face in base metal           | 46 |
| 36. | Figure of a fish in gold                              | 46 |
| 37. | Large figure of a frog in base metal plated with gold | 47 |
| 38. | Small figure of a frog in base metal plated with gold | 47 |
| 39. | Figure of an alligator in gold                        | 48 |
| 40. | Animal figure in base metal plated with gold          | 48 |
| 41. | Bronze bells plated or washed with gold               | 50 |
| 42. | Bronze bell with human features                       | 50 |
| 43. | Triple bell or rattle found on the Rio Grande         | 51 |
| 44. | Ancient Mexican bell                                  | 51 |
| 45. | Fundamental forms of vases—convex outlines            | 58 |
| 46. | Fundamental forms of vases—angular outlines           | 59 |
| 47. | Vases of complex outlines—exceptional forms           | 59 |
| 48. | Vases of compound forms                               | 59 |
| 49. | Square lipped vessel                                  | 59 |
| 50. | Variations in the forms of necks and rims             | 60 |
| 51. | Arrangement of handles                                | 60 |
| 52. | Types of annular bases or feet                        | 61 |
| 53. | Forms of legs                                         | 61 |
| 54. | Grotesque figure forming the handle of a small vase   | 63 |
| 55. | Grotesque figure forming the handle of a small vase   | 63 |

|     |                                                          |    |
|-----|----------------------------------------------------------|----|
| 56. | Grotesque figure forming the handle of a small vase      | 63 |
| 57. | Monstrous figure with serpent shaped extremities         | 63 |
| 58. | Monstrous figure with serpent shaped extremities         | 63 |
| 59. | Grotesque figure                                         | 64 |
| 60. | Grotesque figure                                         | 64 |
| 61. | Grotesque figure                                         | 64 |
| 62. | Figure of a monkey                                       | 64 |
| 63. | Figure of a monkey                                       | 64 |
| 64. | Figure of a monkey                                       | 64 |
| 65. | Animal forms exhibiting long proboscis                   | 65 |
| 66. | Vase illustrating ornamental use of animal figures       | 65 |
| 67. | Vase illustrating ornamental use of animal figures       | 65 |
| 68. | Vase illustrating ornamental use of animal figures       | 66 |
| 69. | Vase illustrating ornamental use of animal figures       | 66 |
| 70. | Series of bowls and cups of unpainted ware               | 67 |
| 71. | Vase of graceful form                                    | 68 |
| 72. | Vase of graceful form                                    | 68 |
| 73. | Vase of fine form ornamented with grotesque heads        | 68 |
| 74. | Vase of fine form ornamented with grotesque heads        | 69 |
| 75. | Vase with ornament of applied nodes and fillets          | 69 |
| 76. | Vase with mantle covered with incised figures            | 70 |
| 77. | Vase with frieze of grotesque heads                      | 70 |
| 78. | Vases with flaring rims and varied ornament              | 71 |
| 79. | Vases with complex outlines and varied ornament          | 71 |
| 80. | Large vase with two mouths and neatly decorated<br>necks | 72 |
| 81. | Large vase with high handles                             | 72 |
| 82. | Top view of high handled vase                            | 73 |
| 83. | Handled vase                                             | 73 |
| 84. | Handled vase                                             | 73 |
| 85. | Handled vase                                             | 73 |

|      |                                                                |    |
|------|----------------------------------------------------------------|----|
| 86.  | Small cup with single handle, ornamented with grotesque figure | 74 |
| 87.  | Small cup with single handle, ornamented with grotesque figure | 74 |
| 88.  | Vase of eccentric form                                         | 74 |
| 89.  | Vessel illustrating forms of legs                              | 75 |
| 90.  | Vessel illustrating forms of legs                              | 75 |
| 91.  | Vessel with large legs decorated with stellar punctures        | 75 |
| 92.  | Vases of varied form with plain and animal shaped legs         | 75 |
| 93.  | Large vase of striking shape                                   | 76 |
| 94.  | Cup with legs imitating animal forms                           | 76 |
| 95.  | Cup with legs imitating a grotesque animal form                | 77 |
| 96.  | Cup with legs imitating the armadillo                          | 77 |
| 97.  | Cup with legs imitating the armadillo                          | 77 |
| 98.  | Cup with frog shaped legs                                      | 77 |
| 99.  | Cup with legs imitating an animal and its young                | 77 |
| 100. | Cups supported by grotesque heads                              | 77 |
| 101. | Large cup supported by two grotesque figures                   | 78 |
| 102. | Cup with two animal heads attached to the sides                | 78 |
| 103. | Cup with two animal heads attached to the sides                | 78 |
| 104. | Vase imitating an animal form                                  | 79 |
| 105. | Vase imitating an animal form                                  | 79 |
| 106. | Vase imitating an animal form                                  | 79 |
| 107. | Fish shaped vessel                                             | 79 |
| 108. | Top view of a fish shaped vessel                               | 80 |
| 109. | Cup with grotesque head attached to the rim                    | 80 |
| 110. | Black cup with incised reptilian figures                       | 81 |
| 111. | Black cup with incised reptilian figures                       | 81 |
| 112. | Black vase with conventional incised pattern                   | 81 |
| 113. | Small cup with conventional incised pattern                    | 82 |

|      |                                                                     |    |
|------|---------------------------------------------------------------------|----|
| 114. | Small tripod cup with upright walls                                 | 82 |
| 115. | Vase with flaring rim and legs imitating animal heads               | 82 |
| 116. | Vase modeled to represent the head of an animal                     | 83 |
| 117. | Pattern upon the back of the vase                                   | 83 |
| 118. | Tripod bowl of red scarified ware                                   | 87 |
| 119. | Tripod bowl of red scarified ware                                   | 87 |
| 120. | Oblong basin with scarified design                                  | 88 |
| 121. | Large scarified bowl with handles imitating animal heads            | 88 |
| 122. | Jar with flat bottom and vertical bands of incised ornament         | 89 |
| 123. | Vase with stand and vertical incised bands                          | 89 |
| 124. | Vase with handles, legs, and vertical ribs                          | 89 |
| 125. | Tripod with owl-like heads at insertion of legs                     | 90 |
| 126. | Tripod with legs rudely suggesting animal forms                     | 90 |
| 127. | Heavy red vase with four mouths                                     | 90 |
| 128. | Vase with horizontally placed handles and rude designs in red       | 91 |
| 129. | Unpolished vase with heavy handles and coated with soot             | 92 |
| 130. | Round bodied vase with unique handles and incised ornament          | 92 |
| 131. | Vase with grotesque figures attached to the handles                 | 93 |
| 132. | Vase with upright handles and winged lip                            | 93 |
| 133. | Top view of vase with winged lip                                    | 94 |
| 134. | Vase with grotesque animal shaped handles                           | 94 |
| 135. | Vase with handles representing strange animals                      | 95 |
| 136. | Vase with handles representing grotesque figures                    | 95 |
| 137. | Vase with handles representing animal heads                         | 96 |
| 138. | Vase with arched handles embellished with life forms in high relief | 96 |

|      |                                                                     |     |
|------|---------------------------------------------------------------------|-----|
| 139. | Vase with arched handles embellished with life forms in high relief | 97  |
| 140. | Tripod vase with shallow basin and eccentric handles                | 99  |
| 141. | Tripod vase with shallow basin and eccentric handles                | 99  |
| 142. | Tripod vase with shallow basin and eccentric handles                | 99  |
| 143. | Tripod vase of graceful shape and neat finish                       | 100 |
| 144. | Heavy tripod vase with widely spreading feet                        | 100 |
| 145. | Neatly modeled vase embellished with life forms and devices in red  | 101 |
| 146. | High tripod vase with incised designs and rude figures in red       | 101 |
| 147. | Handsome tripod vase with scroll ornament                           | 102 |
| 148. | Vase with lizard shaped legs                                        | 102 |
| 149. | Vase with scroll ornament                                           | 103 |
| 150. | Large vase with flaring rim and wide spreading legs                 | 103 |
| 151. | Fragment of a tripod vase embellished with figure of an alligator   | 104 |
| 152. | Vase supported by grotesque human figures                           | 105 |
| 153. | Round bodied vase embellished with figures of monsters              | 106 |
| 154. | Cup with incurved rim and life form ornamentation                   | 107 |
| 155. | Cup with widely expanded rim and constricted neck                   | 107 |
| 156. | Small tripod cup with animal features in high relief                | 108 |
| 157. | Handsome vase supported by three grotesque figures                  | 108 |
| 158. | Vase decorated with figures of frogs and devices in red             | 110 |
| 159. | Vase of unique shape and life form ornamentation                    | 110 |
| 160. | Two-handled vase with life form and linear decoration               | 110 |

|       |                                                                   |     |
|-------|-------------------------------------------------------------------|-----|
| 161.  | Small tripod vase with animal figures in white                    | 111 |
| 162.  | Shapely vase with designs in white paint                          | 112 |
| 163.  | Small red bottle with horizontal bands of ornament                | 115 |
| 164.  | Small red bottle with encircling geometric devices                | 115 |
| 165.  | Bottle with zone occupied by geometric devices                    | 116 |
| 166.  | Bottle with broad zone containing geometric figures               | 116 |
| 167.  | Bottle with decoration of meandered lines                         | 117 |
| 168.  | Bottle with arched panels and geometric devices                   | 117 |
| 169.  | Bottle with arched panels and elaborate devices                   | 118 |
| 170.  | Vase with rosette-like panels                                     | 118 |
| 170a. | Ornament from preceding vase                                      | 118 |
| 171.  | Vase with rosette-like panels                                     | 119 |
| 172.  | Vase with rosette-like panels                                     | 119 |
| 173.  | Theoretical origin of the arched panels                           | 120 |
| 174.  | Theoretical origin of the arched panels                           | 120 |
| 175.  | Theoretical origin of the arched panels                           | 120 |
| 176.  | Vase decorated with conventional figures of<br>alligators         | 120 |
| 177.  | Portion of decorated zone illustrating treatment of<br>life forms | 121 |
| 178.  | Portion of decorated zone illustrating treatment of<br>life forms | 121 |
| 179.  | Vase decorated with highly conventional life forms                | 121 |
| 179a. | Design from preceding vase                                        | 122 |
| 180.  | Vase decorated with highly conventional life forms                | 122 |
| 181.  | Vase decorated with highly conventional life forms                | 123 |
| 182.  | Decorated panel with devices resembling vegetal<br>growths        | 124 |
| 183.  | Vase of unusual shape                                             | 124 |
| 184.  | Vase of unusual shape                                             | 124 |
| 185.  | Vase of unusual shape                                             | 124 |
| 186.  | Double vessel with high arched handle                             | 125 |

|      |                                                                           |     |
|------|---------------------------------------------------------------------------|-----|
| 187. | Double vessel with arched handle                                          | 125 |
| 188. | Vase embellished with life forms in color and in relief                   | 126 |
| 189. | Vase modeled to represent a peccary                                       | 127 |
| 190. | Under surface of peccary vase                                             | 127 |
| 191. | Small vessel with human figures in high relief                            | 127 |
| 192. | Tripod cup with figures of the alligator                                  | 128 |
| 193. | Large shallow tripod vase with geometric decoration                       | 129 |
| 194. | Large bottle shaped vase with high tripod and alligator design            | 130 |
| 195. | Large bottle with narrow zone containing figures of the alligator         | 132 |
| 196. | Vase with decorated zone containing four arched panels                    | 133 |
| 197. | Vase with four round nodes upon which are painted animal devices          | 133 |
| 198. | Vases of varied form and decoration                                       | 134 |
| 199. | Alligator vase with conventional markings                                 | 135 |
| 200. | Alligator vase with figures of the alligator painted on the sides         | 135 |
| 201. | Vase with serpent ornamentation                                           | 136 |
| 202. | Vase representing a puma with alligator figures painted on sides          | 137 |
| 203. | Shallow vase with reptilian features in relief and in color               | 137 |
| 204. | Vase with funnel shaped mouth                                             | 138 |
| 205. | Top view of vase in Fig. 204                                              | 139 |
| 206. | End view of vase in Fig. 204                                              | 139 |
| 207. | Large vase with decorations in red and black                              | 140 |
| 208. | Devices of the decorated zone of vase in Fig. 207, viewed from above      | 141 |
| 209. | Handsome vase with four handles and decorations in black, red, and purple | 142 |

|      |                                                                          |     |
|------|--------------------------------------------------------------------------|-----|
| 210. | Painted design of vase in Fig. 209, viewed from above                    | 143 |
| 211. | Vase of unusual shape, with decoration in black, red, and purple         | 144 |
| 212. | Ornament occupying the interior surface of the basin of vase in Fig. 211 | 144 |
| 213. | Large vase of fine shape and simple decorations                          | 145 |
| 214. | Vase with extraordinary decorative designs                               | 146 |
| 215. | Painted design of vase in Fig. 214, viewed from above                    | 147 |
| 216. | Vase of unique form and decoration                                       | 148 |
| 217. | Painted design of vase in Fig. 216                                       | 148 |
| 218. | Spindle whorl with annular nodes                                         | 149 |
| 219. | Spindle whorl decorated with animal figures                              | 149 |
| 220. | Spindle whorl with perforations and incised ornament                     | 149 |
| 221. | Needlecase                                                               | 150 |
| 222. | Needlecase                                                               | 150 |
| 223. | Needlecase with painted geometric ornament                               | 151 |
| 224. | Needlecase with incised geometric ornament                               | 151 |
| 225. | Needlecase with incised geometric ornament                               | 151 |
| 226. | Statuette                                                                | 152 |
| 227. | Statuette                                                                | 152 |
| 228. | Statuette                                                                | 152 |
| 229. | Statuette                                                                | 152 |
| 230. | Stool of plain terra cotta                                               | 154 |
| 231. | Stool of plain clay, with grotesque figures                              | 155 |
| 232. | Stool of plain terra cotta                                               | 155 |
| 233. | Rattle                                                                   | 157 |
| 234. | Section of rattle                                                        | 157 |
| 235. | Rattle with grotesque figures                                            | 157 |
| 236. | Drum of gray unpainted clay                                              | 158 |

|      |                                                   |     |
|------|---------------------------------------------------|-----|
| 237. | Drum with painted ornament                        | 159 |
| 238. | Painted design of drum in Fig. 237                | 159 |
| 239. | Double whistle                                    | 161 |
| 240. | Section of double whistle                         | 161 |
| 241. | Tubular instrument with two finger holes          | 162 |
| 242. | Section of whistle                                | 162 |
| 243. | Small animal shaped whistle                       | 162 |
| 244. | Small animal shaped whistle                       | 162 |
| 245. | Top shaped whistle                                | 163 |
| 246. | Section, top, and bottom views of whistle         | 164 |
| 247. | Drum shaped whistle                               | 165 |
| 248. | Vase shaped whistle                               | 165 |
| 249. | Crab shaped whistle                               | 166 |
| 250. | Alligator shaped whistle                          | 166 |
| 251. | Cat shaped whistle                                | 167 |
| 252. | Whistle with four ocelot-like heads               | 168 |
| 253. | Bird shaped whistle                               | 169 |
| 254. | Bird shaped whistle                               | 169 |
| 255. | Bird shaped whistle                               | 170 |
| 256. | Whistle in grotesque life form                    | 170 |
| 257. | Conventional figure of the alligator              | 173 |
| 258. | Conventional figure of the alligator              | 173 |
| 259. | Conventional figure of the alligator              | 174 |
| 260. | Conventional figure of the alligator              | 174 |
| 261. | Conventional figure of the alligator              | 174 |
| 262. | Two-headed form of the alligator                  | 175 |
| 263. | Figure of the alligator much simplified           | 175 |
| 264. | The alligator much modified by ceramic influences | 176 |
| 265. | Conventional figure derived from the alligator    | 176 |
| 266. | Conventional figure derived from the alligator    | 176 |
| 267. | Conventional figure derived from the alligator    | 176 |
| 268. | Conventional figure derived from the alligator    | 177 |

|      |                                                        |     |
|------|--------------------------------------------------------|-----|
| 269. | Conventional figure derived from the alligator         | 177 |
| 270. | Conventional figure derived from the alligator         | 177 |
| 271. | Conventional figure derived from the alligator         | 178 |
| 272. | Conventional figure derived from the alligator         | 178 |
| 273. | Conventional figure derived from the alligator         | 178 |
| 274. | Conventional figures derived from the alligator        | 179 |
| 275. | Conventional figure derived from the alligator         | 179 |
| 276. | Conventional figure derived from the alligator         | 180 |
| 277. | Conventional figures derived from the alligator        | 180 |
| 278. | Conventional figures derived from the alligator        | 181 |
| 279. | Conventional figures derived from the alligator        | 182 |
| 280. | Conventional figures derived from the alligator        | 182 |
| 281. | Conventional figures derived from the alligator        | 182 |
| 282. | Conventional figures derived from the alligator        | 182 |
| 283. | Conventional figures derived from the alligator        | 183 |
| 284. | Vase with decorated zone containing remarkable devices | 185 |
| 285. | Series of devices                                      | 185 |
| 286. | Mat or tray with esthetic attributes of form           | 197 |
| 287. | Tray having decided esthetic attributes of form        | 198 |
| 288. | Pyriform water vessel                                  | 198 |
| 289. | Basket with esthetic characters of form                | 199 |
| 290. | Basket of eccentric form                               | 200 |
| 291. | Character of surface in the simplest form of weaving   | 204 |
| 292. | Surface produced by impacting                          | 204 |
| 293. | Surface produced by use of wide fillets                | 204 |
| 294. | Basket with ribbed surface                             | 205 |
| 295. | Bottle showing obliquely ribbed surface                | 205 |
| 296. | Tray showing radial ribs                               | 205 |
| 297. | Combination giving herring bone effect                 | 206 |
| 298. | Combination giving triangular figures                  | 206 |
| 299. | Peruvian work basket                                   | 206 |

|      |                                                                            |     |
|------|----------------------------------------------------------------------------|-----|
| 300. | Basket of Seminole workmanship                                             | 207 |
| 301. | Surface effect produced in open twined combination                         | 207 |
| 302. | Surface effect produced in open twined combination                         | 207 |
| 303. | Surface effect produced by impacting in twined combination                 | 208 |
| 304. | Surface effect produced by impacting the web strands in twined combination | 208 |
| 305. | Surface effect produced by crossing the web series in open twined work     | 208 |
| 306. | Tray with open mesh, twined combination                                    | 208 |
| 307. | Conical basket, twined combination                                         | 209 |
| 308. | Example of primitive reticulated weaving                                   | 210 |
| 309. | Simple form of reticulation                                                | 211 |
| 310. | Reticulated pattern in cotton cloth                                        | 211 |
| 311. | Peruvian embroidery                                                        | 212 |
| 312. | Basket with pendent ornaments                                              | 213 |
| 313. | Basket with pendent ornaments                                              | 213 |
| 314. | Tasseled Peruvian mantle                                                   | 214 |
| 315. | Pattern produced by interlacing strands of different colors                | 216 |
| 316. | Pattern produced by interlacing strands of different colors                | 216 |
| 317. | Pattern produced by interlacing strands of different colors                | 216 |
| 318. | Pattern produced by interlacing strands of different colors                | 217 |
| 319. | Base of coiled basket                                                      | 218 |
| 320. | Coiled basket with geometric ornament                                      | 218 |
| 321. | Coiled basket with geometric ornament                                      | 219 |
| 322. | Coiled basket with geometric ornament                                      | 220 |
| 323. | Coiled basket with geometric ornament                                      | 220 |
| 324. | Coiled basket with geometric ornament                                      | 221 |
| 325. | Coiled basket with geometric ornament                                      | 223 |

|      |                                                                     |     |
|------|---------------------------------------------------------------------|-----|
| 326. | Coiled tray with geometric ornament                                 | 224 |
| 327. | Coiled tray with geometric ornament                                 | 225 |
| 328. | Tray with geometric ornament                                        | 225 |
| 329. | Tray with geometric ornament                                        | 226 |
| 330. | Ornament produced by wrapping the strands                           | 227 |
| 331. | Ornament produced by fixing strands to the surface<br>of the fabric | 227 |
| 332. | Basket with feather ornamentation                                   | 227 |
| 333. | Basket with feather ornamentation                                   | 227 |
| 334. | Piece of cloth showing use of supplementary warp<br>and woof        | 228 |
| 335. | Piece of cloth showing use of supplementary warp<br>and woof        | 228 |
| 336. | Example of grass embroidery                                         | 230 |
| 337. | Example of feather embroidery                                       | 231 |
| 338. | Figures from the Penn wampum belt                                   | 233 |
| 339. | Figures from a California Indian basket                             | 234 |
| 340. | California Indian basket                                            | 234 |
| 341. | Figures from a Peruvian basket                                      | 235 |
| 342. | Figure from a piece of Peruvian gobelins                            | 236 |
| 343. | Figures from a Peruvian vase                                        | 237 |
| 344. | Figure from a circular basket                                       | 238 |
| 345. | Figure of a bird from a Zuñi shield                                 | 239 |
| 346. | Figure of a bird woven in a tray                                    | 240 |
| 347. | Figure of a bird woven in a basket                                  | 241 |
| 348. | Figures embroidered on a cotton net by the ancient<br>Peruvians     | 242 |
| 349. | Figures of birds embroidered by the ancient<br>Peruvians            | 243 |
| 350. | Conventional design painted upon cotton cloth                       | 243 |
| 351. | Herring bone and checker patterns produced in<br>weaving            | 246 |

|      |                                                                          |     |
|------|--------------------------------------------------------------------------|-----|
| 352. | Herring bone and checker patterns engraved in clay                       | 246 |
| 353. | Earthen vase with textile ornament                                       | 247 |
| 354. | Example of textile ornament painted upon pottery                         | 248 |
| 355. | Textile pattern transferred to pottery through costume                   | 248 |
| 356. | Ceremonial adz with carved ornament of textile character                 | 250 |
| 357. | Figures upon a tapa stamp                                                | 251 |
| 358. | Design in stucco exhibiting textile characters                           | 251 |
| 359. | Line of day and numeral symbols from Plates 36c and 37c, Dresden Codex   | 272 |
| 360. | Line of day and numeral characters from Plates 33-39, Dresden Codex      | 276 |
| 361. | Unusual symbol for Akbal from Plate 8 of the Dresden Codex               | 284 |
| 362. | Copy of Plate 50, Dresden Codex                                          | 297 |
| 363. | Copy of Plate 51, Dresden Codex                                          | 306 |
| 364. | Copy of Plate 52, Dresden Codex                                          | 307 |
| 365. | Copy of Plate 53, Dresden Codex                                          | 308 |
| 366. | Copy of Plate 54, Dresden Codex                                          | 309 |
| 367. | Copy of Plate 55, Dresden Codex                                          | 310 |
| 368. | Copy of Plate 56, Dresden Codex                                          | 311 |
| 369. | Copy of Plate 57, Dresden Codex                                          | 312 |
| 370. | Copy of Plate 58, Dresden Codex                                          | 313 |
| 371. | Specimens of ornamental loops from page 72, Dresden Codex                | 337 |
| 372. | Numeral character from the lower division of Plate XV, Manuscript Troano | 343 |
| 373. | Turtle from the Cortesian Codex, Plate 17                                | 348 |
| 374. | Jar from the Cortesian Codex, Plate 27                                   | 349 |
| 375. | Worm and plant from Manuscript Troano, Plate XXIX                        | 351 |
| 376. | Figure of a woman from the Dresden Codex                                 | 351 |

|      |                                                                               |     |
|------|-------------------------------------------------------------------------------|-----|
| 377. | Copy of middle and lower divisions of Plate XIX,<br>Manuscript Troano         | 352 |
| 378. | Copy of lower division of Plate 65, Dresden Codex                             | 353 |
| 379. | The moo or ara from Plate 16, Dresden Codex                                   | 355 |
| 380. | The god Ekchuah, after the Troano and Cortesian<br>Codices                    | 358 |
| 381. | The long nosed god (Kukulcan) or god with the<br>snake-like tongue            | 359 |
| 382. | Copy of head from the Borgian Codex<br>(Quetzalcoatl?)                        | 360 |
| 383. | The supposed god of death from the Dresden Codex                              | 361 |
| 384. | The supposed god of death from the Troano Codex                               | 361 |
| 385. | The god with the banded face from the Troano<br>Codex                         | 362 |
| 386. | The god with the old man's face                                               | 363 |
| 387. | The god with face crossed by lines                                            | 364 |
| 388. | Wooden idol in vessel with basket cover                                       | 371 |
| 389. | Symbolic chart of the Osage                                                   | 378 |
| 390. | Harpoon from Alaska                                                           | 472 |
| 391. | Modern or sealing harpoon                                                     | 472 |
| 392. | Old style naulang or harpoon head                                             | 473 |
| 393. | Modern naulang or harpoon head                                                | 473 |
| 394. | Qilertuang or leather strap and clasps for holding<br>coiled up harpoon lines | 474 |
| 395. | Siatko or harpoon head of the Iglulirmiut                                     | 475 |
| 396. | Siatko found at Exeter Sound                                                  | 475 |
| 397. | Eskimo in the act of striking a seal                                          | 476 |
| 398. | Tutareang or buckle                                                           | 477 |
| 399. | Eskimo awaiting return of seal to blowhole                                    | 478 |
| 400. | Tuputang or ivory plugs for closing wounds                                    | 479 |
| 401. | Wooden case for plugs                                                         | 480 |
| 402. | Another form of plug                                                          | 480 |

|      |                                                               |     |
|------|---------------------------------------------------------------|-----|
| 403. | Qanging, for fastening thong to jaw of seal                   | 480 |
| 404. | Qanging in form of a seal                                     | 480 |
| 405. | Qanging in form of a button                                   | 481 |
| 406. | Qanging serving for both toggle and handle                    | 481 |
| 407. | Qidjarung or whirl for harpoon line                           | 481 |
| 408. | Simple form of whirl                                          | 481 |
| 409. | Old pattern of hook for drawing out captured seal             | 483 |
| 410. | Seal hook of bear's claw                                      | 483 |
| 411. | Modern form of seal hook                                      | 483 |
| 412. | Eskimo approaching seal                                       | 484 |
| 413. | Frame of kayak or hunting boat                                | 486 |
| 414. | Kayak with covering of skin                                   | 487 |
| 415. | Model of a Repulse Bay kayak                                  | 487 |
| 416. | Sirmijaung or scraper for kayak                               | 488 |
| 417. | Large kayak harpoon for seal and walrus                       | 488 |
| 418. | Tikagung or support for the hand                              | 488 |
| 419. | Qatirn or ivory head of harpoon                               | 489 |
| 420. | Manner of attaching the two principal parts of the harpoon    | 489 |
| 421. | Tokang or harpoon head in sheath                              | 489 |
| 422. | Tokang or harpoon head taken from a whale in Cumberland Sound | 490 |
| 423. | Ancient tokang or harpoon head                                | 491 |
| 424. | Teliqbing, which is fastened to harpoon line                  | 492 |
| 425. | Qatilik or spear                                              | 492 |
| 426. | Avautang or sealskin float                                    | 492 |
| 427. | Different styles of poviutang or pipe for inflating the float | 493 |
| 428. | Agdliaq or spear for small seals                              | 494 |
| 429. | Agdliaq points                                                | 494 |
| 430. | Spearheads                                                    | 495 |
| 431. | Large spear head                                              | 495 |

|      |                                                                     |     |
|------|---------------------------------------------------------------------|-----|
| 432. | Anguvigang or lance                                                 | 496 |
| 433. | Nuirn or bird spear                                                 | 496 |
| 434. | Nuqsang or throwing board                                           | 496 |
| 435. | Sealing at the edge of the ice                                      | 498 |
| 436. | Model of sakurpāng or whaling harpoon                               | 500 |
| 437. | Niutang with floats                                                 | 500 |
| 438. | Wooden bow from Iglulik                                             | 502 |
| 439. | Wooden bow from Cumberland Sound                                    | 502 |
| 440. | Bows of reindeer antlers                                            | 503 |
| 441. | Bow of antlers with central part cut off straight                   | 503 |
| 442. | Arrows with bone heads                                              | 504 |
| 443. | Arrows with metal heads                                             | 504 |
| 444. | Arrow head                                                          | 505 |
| 445. | Showing attachment of arrowhead vertically and<br>parallel to shank | 505 |
| 446. | Various forms of arrowhead                                          | 506 |
| 447. | Socket of spear handle from Alaska                                  | 506 |
| 448. | Slate arrowhead                                                     | 506 |
| 449. | Flint arrowheads from old graves                                    | 507 |
| 450. | Various styles of quiver                                            | 507 |
| 451. | Quiver handles                                                      | 508 |
| 452. | Whalebone nooses for catching waterfowl                             | 511 |
| 453. | Kakivang or salmon spear                                            | 512 |
| 454. | Ivory fish used as bait in spearing salmon                          | 513 |
| 455. | Quqartaun for stringing salmon                                      | 514 |
| 456. | Salmon hook                                                         | 515 |
| 457. | Salmon hook                                                         | 515 |
| 458. | Bait used in fishing with hooks                                     | 516 |
| 459. | Butcher's knife with bone handle                                    | 516 |
| 460. | Pana or knife for dissecting game                                   | 517 |
| 461. | Form of ulo now in use                                              | 518 |
| 462. | Old ulo handle from Cape Broughton, Davis Strait                    | 518 |

|      |                                                       |     |
|------|-------------------------------------------------------|-----|
| 463. | Fragment of an ulo blade made of slate                | 518 |
| 464. | Ulo handle from recent grave                          | 518 |
| 465. | Modern tesirqun or scraper                            | 519 |
| 466. | Old style tesirqun or scraper                         | 519 |
| 467. | Seligoung or scraper used for softening skins         | 520 |
| 468. | Old stone scrapers found in graves                    | 521 |
| 469. | Stretcher for lines                                   | 522 |
| 470. | Ivory needle                                          | 523 |
| 471. | Ivory needlecase from Cumberland Sound                | 523 |
| 472. | Common pattern of needlecase                          | 523 |
| 473. | Tikiq or thimble                                      | 524 |
| 474. | Instrument for straightening bone                     | 525 |
| 475. | Drill for working in ivory and bone                   | 525 |
| 476. | Driftwood used in kindling fire                       | 526 |
| 477. | Eskimo graver's tool                                  | 526 |
| 478. | Framework of Eskimo boat                              | 527 |
| 479. | Kiglo or post                                         | 527 |
| 480. | Umiaq or skin boat                                    | 528 |
| 481. | Umiaq or skin boat                                    | 528 |
| 482. | Qamuting or sledge                                    | 529 |
| 483. | Sledge shoe                                           | 530 |
| 484. | Form of clasp for fastening traces to sledge          | 531 |
| 485. | Artistic form of clasp for fastening traces to sledge | 531 |
| 486. | Uqsirn for fastening traces to pitu                   | 532 |
| 487. | Ano or dog harness                                    | 532 |
| 488. | Sadniriah or clasp                                    | 532 |
| 489. | Tube for drinking                                     | 535 |
| 490. | Various styles of snow knife                          | 539 |
| 491. | Ground plan of snow house of Davis Strait tribes      | 540 |
| 492. | Snow house of Davis Strait, sections                  | 541 |
| 493. | Section and interior of snow house                    | 543 |
| 494. | Ukusik or soapstone kettle                            | 545 |

|      |                                                                      |     |
|------|----------------------------------------------------------------------|-----|
| 495. | Plan of double snow house                                            | 546 |
| 496. | Plan of Iglulik house                                                | 547 |
| 497. | Plan of Hudson Bay house                                             | 547 |
| 498. | Plan and sections of qarmang or stone house                          | 548 |
| 499. | Plan of large qarmang or stone house                                 | 549 |
| 500. | Plan of stone house in Anarnitung, Cumberland<br>Sound               | 549 |
| 501. | Plan of groups of stone houses in Pangnirtung                        | 530 |
| 502. | Plan of qarmang or house made of whale ribs                          | 550 |
| 503. | Storehouse in Ukiadliving                                            | 551 |
| 504. | Plan and sections of tupiq or tent of Cumberland<br>Sound            | 551 |
| 505. | Plan and section of tupiq or tent of Pond Bay                        | 553 |
| 506. | Plan and section of double winter tent, Cumberland<br>Sound          | 553 |
| 507. | Qaturang or boot ornament                                            | 554 |
| 508. | Woman's jacket                                                       | 555 |
| 509. | Ivory beads for women's jackets                                      | 555 |
| 510. | Girdle buckles                                                       | 556 |
| 511. | Infant's clothing                                                    | 557 |
| 512. | Child's clothing                                                     | 557 |
| 513. | Ivory combs                                                          | 559 |
| 514. | Buckles                                                              | 560 |
| 515. | Manner of tattooing face and wearing hair                            | 561 |
| 516. | Manner of tattooing legs and hands                                   | 561 |
| 517. | Forks                                                                | 563 |
| 518. | Ladle of musk ox horn                                                | 563 |
| 519. | Skull used in the game ajegaung                                      | 565 |
| 520. | Ivory carving representing head of fox, used in the<br>game ajegaung | 565 |
| 521. | Ivory carvings representing polar bear, used in the<br>game ajegaung | 566 |

|      |                                                                         |     |
|------|-------------------------------------------------------------------------|-----|
| 522. | Figures used in playing tingmiujang, a game similar to dice             | 567 |
| 523. | Game of nuglutang                                                       | 568 |
| 524. | The sāketān or roulette                                                 | 569 |
| 525. | Ajarorpoq or cat's cradle                                               | 569 |
| 526. | Ball                                                                    | 570 |
| 527. | Dolls in dress of the Oqomiut                                           | 571 |
| 528. | Dolls in dress of the Akunirmiut                                        | 571 |
| 529. | Modern snow goggles of wood                                             | 576 |
| 530. | Old form of snow goggles of ivory                                       | 576 |
| 531. | Diagram showing interior of qaggi or singing house among eastern tribes | 600 |
| 532. | Plan of Hudson Bay qaggi or singing house                               | 601 |
| 533. | Kilaut or drum                                                          | 602 |
| 534. | Plans of remains of supposed qaggi or singing houses                    | 603 |
| 535. | Qailertétang or masked figure                                           | 606 |
| 536. | Model of lamp from a grave in Cumberland Sound                          | 613 |
| 537. | Qaudjaqdjuq is maltreated by his enemies                                | 631 |
| 538. | The man in the moon comes down to help Qaudjaqdjuq                      | 631 |
| 539. | The man in the moon whipping Qaudjaqdjuq                                | 632 |
| 540. | Qaudjaqdjuq has become Qaudjuqdjuaq                                     | 632 |
| 541. | Qaudjuqdjuaq killing his enemies                                        | 633 |
| 542. | Tumiujang or lamp of the Tornit                                         | 634 |
| 543. | Cumberland Sound and Frobisher Bay, drawn by Itu, a Nugumio             | 644 |
| 544. | Cumberland Sound and Frobisher Bay, drawn by Sunapignang, an Oqomio     | 645 |
| 545. | Cumberland Sound, drawn by Itu, a Nugumio                               | 646 |
| 546. | Peninsula of Qivitung, drawn by Angutuqdjuaq, a Padlimio                | 647 |

1. In pocket at end of volume.

## LETTER OF TRANSMITTAL.

SMITHSONIAN INSTITUTION, BUREAU OF ETHNOLOGY,  
*Washington, D. C., October 23, 1885.*

SIR: I have the honor to submit my Sixth Annual Report as Director of the Bureau of Ethnology.

The first part consists of an explanation of the plan and operations of the Bureau; the second part consists of a series of papers on anthropologic subjects, mainly prepared by my assistants to illustrate the methods and results of the work of the Bureau.

I desire to express my thanks for your earnest support and your wise counsel relating to the work under my charge.

I am, with respect, your obedient servant,

A handwritten signature in dark ink, appearing to read "J. M. Powell". The signature is written in a cursive style with a large, looped initial "J" and a long, sweeping underline.

Prof. SPENCER F. BAIRD,  
*Secretary Smithsonian Institution.*

**SIXTH ANNUAL REPORT  
OF THE  
BUREAU OF ETHNOLOGY.**

BY J. W. POWELL, DIRECTOR.

## INTRODUCTION.

The prosecution of research among the North American Indians, as directed by act of Congress, was continued during the fiscal year 1884-'85.

No change has been made in the general plan upon which the work has been prosecuted as set forth in former reports. Certain lines of investigation have been decided upon, which are confided to persons trained in their pursuit, and the results of these labors are presented from time to time in the publications of the Bureau provided for by law. A brief statement of the work upon which each of the special students was actively engaged during the fiscal year is furnished below; this, however, does not embrace all the studies undertaken or services rendered by them, since particular lines of research have been suspended in this, as in former years, in order to prosecute temporarily work regarded as of paramount importance. From this cause delays have been occasioned in the completion of several treatises and monograph's, already partly in type, which otherwise would have been published.

Invitation is renewed for the assistance of explorers, writers, and students who are not and may not desire to be officially connected with the Bureau. Their contributions, whether in the shape of suggestions or of extended communications, will be gratefully acknowledged, and, if published either in the series of reports or in monographs or bulletins, as the liberality of Congress may in future allow, will always receive proper credit.

The items now reported upon are presented in three principal divisions. The first relates to the publication made; the second, to the work prosecuted in the field; and the third, to the office work, which largely consists of the preparation for publication of the results of field work, with the corrections

and additions obtained from the literature of the subjects and by correspondence.

## **PUBLICATION.**

The only publication actually issued during the year was entitled Proof-Sheets of a Bibliography of the Languages of the North American Indians, by James Constantine Pilling. The volume, a quarto of 1,175 pages, consists of an author catalogue of books, manuscripts, magazine and newspaper articles, publications of learned societies, and other documents relating in any way to the Indian languages of North America. Only one hundred copies were printed, which were distributed to collaborators.

This work was commenced by Mr. Pilling in 1879 and has been prosecuted with diligence and skill, notwithstanding the engrossing nature of his other duties. It began as an author card catalogue, designed merely for office use. In time it became apparent that such a systematic catalogue of the literature of Indian languages, if printed and distributed, would be of important service to all the numerous workers on the general subject, besides those directly connected with the Bureau, to whom alone it was accessible in manuscript form. By this course the accumulated results of several years' labor would be immediately available for the use of students generally, and the distribution of proof-sheets would in turn increase interest in the work, elicit comment and criticism, and secure additional contributions, through all of which the final volume contemplated would become more satisfactory and complete, both in form and substance. The thorough conscientiousness and punctilious care shown in the present catalogue, and especially the comprehensive bibliographic spirit in which the work has been conceived, prove the peculiar fitness of the author for the undertaking. He has set before him and has kept steadily in view the following aims:

First, to discover every document in existence relating to the subject, either printed or in manuscript.

Secondly, to record a description of every document found, so accurate and full that each book or article mentioned is clearly identified and all its contents relating to Indian languages set forth, with citation of the chapters and pages within the work where the linguistic material may be found.

Thirdly, to name, when possible, one or more libraries where each work catalogued may be found.

Fourthly, to arrange and combine the whole so that the student using it may in the shortest time learn whether any work contains the special matter which he desires to consult, and, if so, precisely where he may find it. In the case of rare books or papers special attention has been paid to obtain full information, and in the case of some of the rarest books fac-similes of the title pages are given. The value of a work so broadly conceived and so carefully executed is very great. The literature of this subject has become so voluminous, so disconnected, so scattered in time and place, that progress in the classification of Indian languages and the determination of their affinities has been greatly retarded, awaiting the orderly arrangement of accumulated information. This requisite, with the important addition of the correction of current errors, is met by the catalogue. It has been found indispensable to the Bureau and has already been gratefully acknowledged as invaluable by all students of American tribes to whom copies have been distributed.

Since the printing and distribution of the proof-sheets, and markedly as a result thereof, the card catalogue has continued to grow; and, although not complete and, from the nature of the subject, not expected to become absolutely exhaustive, the recent additions to it indicate how thoroughly the work was originally done. It may be possible, therefore, before long to substitute for the Proof-Sheets the Bibliography itself in standard form.

## **FIELD WORK.**

Under this heading are comprised—

First, the systematic operations of the division of mound exploration carried on east of the Rocky Mountains.

Secondly, researches in and collections from the ancient ruins of the Southwest and comparative study of the present inhabitants of that region and the objects found among them.

Thirdly, linguistic work or expeditions among the several Indian tribes at their homes, with the main purpose of acquiring knowledge of their spoken languages.

Fourthly, general studies, or those embracing various branches of inquiry, conducted among the existing Indian tribes.

## **MOUND EXPLORATIONS.**

### **WORK OF PROF. CYRUS THOMAS.**

The work of exploring the mounds and other ancient monuments of that portion of the United States east of the Rocky Mountains, commenced in 1882, was carried on during the fiscal year, under the charge of Prof. Cyrus Thomas.

The regular assistants during the first half of the year were Messrs. P.W. Norris, James D. Middleton, and John P. Rogan. For the latter half they were Messrs. Middleton, Rogan, and John W. Emmert, the last named having been engaged to fill the vacancy occasioned by the death of Mr. Norris.

Mr. Norris was engaged during the fall of 1884 in exploring the extensive group of works in the vicinity of Charleston, Kanawha Valley, W. Va. He continued at work there until December, when he was compelled by cold weather and illness to desist. To the great regret of all his associates in the work, his illness terminated in death on the 14th of January, 1885. By his death the division has lost a faithful and enthusiastic worker.

During the summer and fall of 1884 and until the approach of extremely cold weather, Mr. Middleton was engaged in exploring the works of Knox County, Ohio. Throughout the winter and following spring his field of operations was eastern Arkansas. In the latter field he was assisted by Mr. L. H. Thing, who was employed for three months as temporary assistant.

During the summer and until the beginning of winter, Mr. Rogan was engaged (in conjunction with Rev. J. P. Maclean, who was employed as a temporary assistant) in exploring the ancient monuments of Butler County and the adjacent regions of southern Ohio. On the approach of the cold season he went south, his field of operations for the remainder of the year being northern Georgia and the southern counties of East Tennessee.

Mr. Emmert, who had been employed on January 1, 1885, to make some special explorations in East Tennessee, was made permanent assistant

immediately after the death of Mr. Norris. His work in that section proving successful he continued it until the close of the fiscal year.

Mr. Gerard Fowke was engaged during November in examining the ancient quarries of Flint Ridge, Ohio, and in making a collection to illustrate the various stages in the aboriginal manufacture of flint implements. His collection is, perhaps, the most complete in this particular line of any so far made in this country. In the winter he was employed about two months in special investigations of some ancient works in Pontotoc and Union Counties, Miss., a locality supposed to have been visited by De Soto during his unfortunate expedition. In some of the mounds of this section, which was formerly the home of the Chikasa, he found some articles of European manufacture, among them a small silver plate bearing the royal arms of Castile and Leon in an old heraldic form.

Although the number of specimens obtained does not exceed that of the collection of the preceding year, the general result shows a decided advance in the accuracy of the work done. The measurements and plats have been made with more care and exactness, the descriptions are more complete, and the details more fully set forth. As an illustration one case is presented. A large mound was opened which was found to contain over ninety skeletons, irregularly placed and at different depths. At the outset a plat of the mound was made; each skeleton was located on it as discovered, and notes were taken of the depth, position, articles found with it, etc. Thus the exact position of each skeleton in the mound is recorded, as well as that of any article accompanying it. The collections made are more varied in character than those of any previous year, including several new types of pottery, some unusually fine stone implements, and from several mounds articles showing contact with Europeans. The pottery obtained by Messrs. Middleton and Thing in Arkansas is of more than ordinary interest, containing a number of specimens of the rarer forms, also several colored specimens.

The same care has been taken as heretofore in labeling and numbering the specimens, so that each can be traced by the record to the exact place where it was found. The illustrations showing the construction, character, and form of the various works explored exceed in number, accuracy, and importance those of any previous year.

## **EXPLORATIONS IN THE SOUTHWEST.**

### **WORK OF MR. JAMES STEVENSON.**

Mr. James Stevenson was placed in charge of a party, with instructions to proceed to Arizona and New Mexico to make researches and collections among the Pueblo Indians and the ancient ruins in that region.

Mr. Stevenson's party was divided into three sections. The section in charge of Mr. F. T. Bickford visited the remarkable series of ruins in Chaco cañon, in northwestern New Mexico; Cañon de Chelly and its branch cañons; the cliff dwellings in Walnut cañon, in Arizona, and a group of interesting cave dwellings, different in structure from any heretofore found, near Flagstaff, in the same Territory. All these were carefully examined. Full and extensive notes, as well as sketches and photographic illustrations, were made of these ruins.

Another section, in charge of Mr. C. A. Garlick, was stationed at the pueblo of Acoma, in New Mexico. The work at this village resulted in a collection of about thirty-five hundred specimens, consisting of pottery and a variety of utensils of other material, such as stone, bone, wood, and woven fabrics, illustrating the arts of the people of Acoma. The collections from this pueblo, though not embracing a great variety of objects, will illustrate nearly all the phases of the arts and industrial pursuits of these Indians.

Another section of Mr. Stevenson's party, under his own supervision and with the important assistance of Mrs. Stevenson, was employed in making collections and studies at Zuñi. The collection from there is much larger than any heretofore obtained and includes many objects relating to the outdoor ceremonies of the Zuñi. Specimens of these were secured from their sacred springs, caves, and shrines. All details relating to their ceremonials were attentively studied, and a series of water color sketches was made of altars used and of masks worn on these important occasions. A large number of fetiches was also obtained, representing many of the animals held in religious esteem by the Zuñi. A series of photographs was made of the sacred springs, wells, monuments, picture writings, and shrines

of the Zuñi located at different points over an area of about seventy-five miles from Zuñi, and a collection was secured of representative specimens of their fetiches, plume sticks, and other objects connected with their mythology and religious practices. The collection made during the year was unusually large and important. It comprises about eighty-five hundred specimens from the Indian tribes of the Southwest embraced in the research; these consist of woven fabrics and pottery, bone, and stone implements, both ancient and modern, and represent nearly all phases of the life, art, and industries of these tribes. These collections have been deposited in the U. S. National Museum for arrangement, classification, and description.

### **WORK OF MR. VICTOR MINDELEFF.**

A party in charge of Mr. Victor Mindeleff left Washington on August 5 to survey the ruined pueblos of the Chaco, in New Mexico. Five of the ruins were accurately measured and platted to scale, and a full series of sketches, plans, and photographs was secured. Mr. Mindeleff returned from the field on the 1st of October. He then made a trip to the great Etowah mound, near Cartersville, Ga., under the direction of Prof. Cyrus Thomas, in order to secure an accurate survey and scale drawing, as a basis for the construction of a model.

At the close of this work Mr. Mindeleff returned to Washington, on October 7, and was engaged in office work until the middle of the following June, when he took the field in advance of his party for further studies among the ruins and pueblos of the Cibola and Tusayan groups. He was also instructed to secure similar material at other available points for comparison.

## **LINGUISTIC FIELD WORK.**

### **WORK OF MRS. ERMINNIE A. SMITH.**

From the 1st of July to the 15th of August, 1884, Mrs. Smith, assisted by Mr. J. N. B. Hewitt, of Tuscarora descent, was engaged among the Onondaga living near Syracuse, N. Y., in translating and annotating two Onondaga manuscripts; afterward, until the latter part of October, with the same assistance, she was at work on the Grand River reservation in Canada, where she filled out the vocabulary in the Introduction to the Study of Indian Languages from the dialect of the Cayuga. She also obtained from the Mohawk a translation, with annotations, of a manuscript in their dialect.

The three manuscripts mentioned are now in the possession of the Bureau of Ethnology. Their origin and history are not distinctly known, as they are all probably copies of originals which seem to have been lost or destroyed. It was intended in these manuscripts to reproduce, by the alphabet and the script used by English writers, the sound of the dialects employed.

These records have their chief interest in the preservation of many archaic words, or those of ceremony, law, and custom, which in these dialects, as is the general rule, remain unchanged, although the colloquial language may be modified. The subject matter of all these records is genuinely and exclusively Iroquoian.

The Mohawk manuscript was copied about the year 1830 by Chief John "Smoke" Johnson from an earlier original or perhaps copy. The orthography of this copy is quite regular and is that of the early English missionaries, being similar in many respects to the well known Pickering alphabet.

One of the Onondaga manuscripts was found in the possession of Mr. Daniel La Fort and the other in that of Mrs. John A. Jones, both of the Onondaga reserve, New York. These two copies differ from each other in orthography and substance, the Jones manuscript being probably a full detail of a part of the other.

The orthography of the La Fort manuscript is very irregular and difficult to read, but that of the Jones manuscript is regular and legible. The Mohawk manuscript contains a detailed account of the rites and ceremonies, speeches and songs, of the condoling and inducting council of the Iroquoian League in the form in which that council was conducted by the elder brothers or members of the Onondaga, Mohawk, and Seneca divisions, which have been generally called tribes, but are more correctly confederacies, their villages being the tribal unit. The La Fort Onondaga manuscript comprises a similar ritual of the same council as carried out by the younger brothers, viz., the Cayuga, Oneida, and Tuscarora members or confederacies of the league. The Jones Onondaga manuscript is the charge of the principal shaman to the newly elected or inducted chief or chiefs.

During the remainder of the year material was collected and work continued on the Tuscarora-English part of the Tuscarora dictionary.

### **WORK OF MR. H. W. HENSHAW.**

Mr. H. W. Henshaw visited southern California for the purpose of pursuing linguistic studies in the group of languages spoken by the Santa Barbara Indians. Although these Indians became known at a very early day, being mentioned with particularity in the relation of Cabrillo's voyage along the California coast in 1542, but little has been ascertained in respect to their language and its relations to the speech of neighboring tribes.

Few vocabularies were collected by the early Spanish missionaries and those gathered were very imperfect, so that no conclusions can be based upon them with confidence.

As a result of the policy pursued by the various missionaries among these docile tribes, aboriginal habits were soon exchanged for others imposed by the priests. Tribal organizations were broken up and the Indians were removed from their homes and located about the missions. In addition the Spanish language was early introduced and so far as possible made to replace the aboriginal tongue. As a consequence Spanish became familiar to a large number of the proselytes, and all the surviving Santa Barbara Indians speak Spanish fluently, or rather the Mexican dialect of Spanish. Indeed, the impression prevails generally in California that none of the

Indians can speak their own tongue. As a matter of fact, however, in their own families and when away from the white men they discard Spanish entirely.

The attempt to preserve the language was begun none too soon, as of the large population attributed to this part of the California coast Mr. Henshaw was able to discover only about fifty survivors, and these were widely scattered over several counties. A number of the dialects of the linguistic family are now extinct, and only a month before Mr. Henshaw's arrival at San Buenaventura an old woman died who, it is believed, was the last person to speak the dialect belonging to the Island of Santa Cruz. In Santa Barbara and Ventura counties six dialects of the family were found, which are believed to be all that are now extant.

In the case of the dialect of Santa Rosa island, but one Indian remained to speak it. Two more dialects are spoken by two or three individuals only. The existing dialects, named according to the missions around which they were spoken, are as follows: San Buenaventura, Santa Barbara, Santa Rosa Island, Purissima, Santa Inez, and San Luis Obispo. With the exception of the last named the several dialects are very closely related, and, although each possesses a greater or less number of words not contained in the others, their vocabularies show many words which are common to all.

The dialect formerly spoken at San Luis Obispo differs much from any of the others, and a critical comparison is necessary to reveal a sufficient number of words possessing identical roots to render their common parentage obvious.

Extensive vocabularies of the dialects of San Antonio and San Miguel were obtained, there being about a dozen Indians who speak these languages around the old San Antonio mission. These languages have been supposed to be of the Santa Barbara family (as it has hitherto been termed, now called Chumashan family), but the material obtained by Mr. Henshaw disproves this, and, for the present at least, they are considered to form a distinct family.

Mr. Henshaw visited Los Angeles and San Diego counties for the purpose of determining the exact northern and southern limits of the Shoshonian family, which extends quite to the coast in California.

At San Diego and San Luis Rey he obtained vocabularies representing four dialects of the Yuman family.

### **WORK OF MR. A. S. GATSCHET.**

In August, 1884, Mr. Gatschet proceeded to visit the Tonkawē and Lipan tribes in Texas.

He reached Fort Griffin on the 29th of August. The Tonkawē tribe was encamped about a mile and a half south of Fort Griffin, Shackleford county, and consisted of 78 individuals, while the Lipan camp, one mile north-northwest, consisted of 19 persons only. All these Indians were on the point of removing to the Oakland reserve, Indian Territory.

The Tonkawē constitute an aggregate of several tribal remnants formerly living independently of one another in southern Texas and on the Rio Grande. Mr. Gatschet devoted five weeks to the study of their language and one week to that of the Lipan, which is a dialect of Apache (Athapaskan). The Tonkawē is a sonorous and energetic form of speech. The radix of many of the adjectives becomes reduplicated to form a kind of plural, and the same thing is observed in some of the verbs, where iteration or frequency has to be indicated. Case suffixes are observed in the substantive, which can easily be traced to postpositions as their original forms. Very few of the natives were sufficiently conversant with English or Spanish to serve as interpreters, so that it was difficult to secure trustworthy results. A white man who had lived over six years among them was of material help, and several mythologic and other texts were obtained with tolerable correctness through his aid.

On October 9 Mr. Gatschet left Fort Griffin and reached Fort Sill, in the Indian Territory, on the 15th. Many Kaiowē and Comanche Indians encamped during the warmer months of the year around this fort, which is situated at the southeast base of the Wichita mountains. He engaged the best help he could find for studying the Kaiowē language, for which there is no Government interpreter. The Comanche is the predominating language on the whole Kaiowē, Comanche, and Apache reservation, although the Comanche exceed the Kaiowē but little in number. The Comanche is more

easily acquired, at least to the extent required in conversation, and all the traders and shopkeepers on the reservation have a smattering of it.

Better interpreters for Kaiowē were obtained at Anadarko, the seat of the agency, where Mr. Gatschet remained from October 31 to December 12. A few Kaiowē were found who had passed some months or years among Americans or at the Indian schools at Carlisle, Chilocco, and elsewhere, and could express themselves intelligibly in English. A few white Mexicans were found among the Comanche, who were captured by them in infancy, acquired the Comanche language, and have ever since lived among these Indians. Of the Kaiowē, Mr. Gatschet acquired over two thousand terms, phrases, and sentences, several historic texts of value, and of the Comanche, eight hundred or a thousand words. The circumstances necessitated careful and numerous revisions of everything obtained, by which much of the time was absorbed.

The Na-ishi Apache, about four hundred in number and formerly roaming with the Kaiowē, furnished also a large amount of terms, exceeding fifteen hundred.

There are a few verbal similarities between the Kaiowē and the Shoshoni languages, but apparently not enough to indicate anything more than long association of these peoples. The Kaiowē has a dual in the intransitive verb and in some nouns. There are more than a dozen different modes of forming the plural of nouns. The subject pronoun is incorporated with the verb as a prefix, and every tense has a different subject pronoun, as in Otomi and other languages of southern Mexico.

Vocabularies were also obtained of Delaware, Ottawa, Yuchi, Caddo, Wichita, and of the hitherto unstudied Caddo dialects of Anadarko and Yatassi.

In spite of persevering search it was not possible to find any of the Bidai or the Tonica in Texas, although it is probable that some of them survived in that State as late as 1850.

Mr. Gatschet then passed a whole month among the Atakapa at Lake Charles, the county seat of Calcasieu parish, Louisiana. Of the two dialects traceable, only the western one seems to exist now, being still spoken by a few women living at the town. The language is sonorous, but strongly nasal.

Returning to the Indian Territory, after a fruitless search for the Tonica and Adai, he stopped at Eufaula, Creek Nation, to meet a Na'htchi Indian named Lasley, about sixty years old, who had represented his tribe in the councils of the Creek Nation. This man explained his Na'htchi terms and phrases by Creek equivalents, and these had to be translated into English to obtain full light concerning the Na'htchi terms. One legendary text was also obtained. The language is rather consonantal and has a multiplicity of verbal forms.

Among the Yuchi tribe on Middle Arkansas river, southwestern bank, and over 40 miles from Muscogee Station, Indian Territory, he remained but a week, too short a time to obtain full information respecting this interesting language. There are five or six hundred Yuchi still living on this tract. Two texts and a few popular songs, with one thousand terms of the language, were obtained.

The last stop was made among the Modoc at Quapaw Agency, at the agency buildings. About ninety are left of those brought there for having taken part in the Modoc war of 1872-'73. Five mythic tales were gathered from the natives within the short time of three weeks, one of them being of considerable length and of importance. It is called "The birth of Aishish." The birth of this astral deity resembles in most particulars that of Bacchus from the thigh of Jupiter after his mother, Semele, had been burned to death. The terms, phrases, and sentences gathered, besides the myth mentioned, amount to over fifteen hundred items, which will prove useful for completing the work on the Klamath Indians of Oregon now in preparation.

Of the Shawnee language several hundred words were gathered from the Indians of that tribe settled around the agency.

Mr. Gatschet returned to Washington in April, 1885.

### **WORK OF REV. J. OWEN DORSEY.**

Rev. J. Owen Dorsey visited the Siletz Agency, Oregon, in August, 1884, to gain linguistic and other information respecting the tribes in that region. When he returned, in November, he brought back as the result of his work the following vocabularies:—Athapascan family: Applegate Creek, Galice

Creek, Chastā Costa, Miko-no-tunne, Chetco, Smith River, Cal., and Upper Coquille.—Yakonan family: Yaquina, Alsea, Siuslaw, and Lower Umpqua.—Kusan family: Mulluk or Lower Coquille.—Takilman family: Takilma or Upper Rogue River.—Shahaptian family: Klikitat.—Sastean family: Shasti—total, nineteen vocabularies, ranging from fifty to three thousand entries, exclusive of phrases and grammatical notes.

He also obtained materials for an account of the social organization into villages of some of these Indians, the basis for which appears to have been the clan or gens. Rough maps, showing the localities of the villages, were made. Mr. Dorsey also obtained from several tribes the corresponding Indian names of about sixty vegetal products, specimens of which were brought to Washington for identification.

### **WORK OF MR. JEREMIAH CURTIN.**

Mr. Curtin spent the first two weeks of July at the Quapaw agency, Indian Territory, in making a collection of Modoc myths, which he had begun in the preceding winter, being part of a general collection of Indian myths begun in 1883. The number of Modoc myths obtained was nearly one hundred.

After finishing work at the Quapaw Agency, he returned to Washington, and shortly afterward was directed to proceed to northern California and obtain vocabularies of the Nosa and Kombo languages, and thence to Oregon to obtain vocabularies of the Wasco, Tyigh, and Tenina languages.

Work was begun on the Nosa language (Yanan family) at Redding, Cal., on October 11. The difficulties were very great, especially at first, owing to the fact that the Nosa are few in number, live far from one another, and have a very imperfect knowledge of English.

The Nosa were a prominent and rather numerous people until 1864, when all of them who could be found were massacred by white settlers, who organized two companies for the purpose of exterminating the tribe. Owing to a chance by which a few escaped and to the exertions of Mr. Benjamin Oliver, who secreted several in his cellar, about fifteen full blood Nosa survived.

Work on Nosa was continued in and around Redding until the end of November, when Round Mountain was visited to complete the Nosa vocabulary and obtain that of the Atsugei (Palaikan family), a very interesting language. Work at Round Mountain was finished on January 8 and Redding was revisited on January 9, preparatory to departing for Oregon.

Owing to the excessive severity of the winter and the snow blockades, which lasted six weeks, communication with Warm Spring was closed, and it was impossible to enter the reservation till January 27, when Sinnashee, a school and center of the Warm Spring Indian population, was reached.

At this place the Tyigh vocabulary (Shahaptian family) was collected. The Wasco (Chinookan family) was obtained at the agency headquarters near the Deschutes river. Tenina, being identical with the Tyigh language, was omitted. From April 18, at which date work at the Warm Spring agency was finished, until June 30, the time was devoted to collecting myths in the Klamath reservation and at Yreka.

During the whole period of work all the myths that could be found among the people whose languages were being investigated were reduced to writing. In this manner a large body of Nosa, Atsugei, Tyigh, and Wasco myths was collected. In the cases of Klamath and Shasti, myths were the objects directly in view.

The vocabularies were obtained with satisfactory completeness and the verbal systems worked out in detail.

The Nosa is remarkable for a regularity of structure which yields to analysis and has a certain monotonous harmony of sound.

The Atsugei has a sonorous roll, a strong letter *r*, and a certain number of words in common with the Shasti, itself one of the *r* languages.

## **GENERAL FIELD WORK.**

### **WORK OF DR. WASHINGTON MATTHEWS.**

Dr. Washington Matthews, assistant surgeon U. S. Army, continued his investigations among the Navajo Indians in New Mexico and Arizona. He had been stationed in the Navajo country as post surgeon of Fort Wingate, N. Mex., from 1880 to 1884, during which time he devoted himself to studying the language, customs, and ceremonies of this tribe as much as his official duties would permit. Some of the great shamanistic ceremonies of the Navajo, occupying nine days for their performance, he had often seen in part; but he had never had an opportunity of witnessing one throughout its entire duration, as he had not sufficient time at his disposal.

Before leaving New Mexico, however, he secured the friendship and confidence of some of the leading medicine men and obtained their promise to admit him to their most secret rites during their entire performance whenever he should be able to avail himself of the privilege. He was also promised complete instruction in the mythology and symbolism of these rites.

In the autumn of 1884 he was given an opportunity, under the auspices of the Bureau of Ethnology, to return to the Navajo country and devote himself for a considerable time entirely to anthropologic studies among the people.

He first visited the Navajo who dwell in the neighborhood of the San Mateo mountains, the Tsotsildinè, or people of the Great Peak, a local division or subtribe living much farther to the east and having longer and more intimate associations with Mexicans and Americans than the main body of the people. While at this place, he ascended the peak of San Mateo, or Mount Taylor, a mountain held sacred by the Navajo, to observe the various places on the mountain mentioned in the Navajo myths.

Leaving San Mateo he proceeded to Fort Wingate, and learning that one of the most important of the Navajo rites was about to be celebrated at a place called Niqotlizi (Hard Earth), north of Fort Wingate on the Navajo

reservation, he repaired thither without delay. The ceremony which he went to witness was that of *dsilyídje-qaçàl*, or mountain chant. It is also called *Ilñasjingo-qaçàl*, or chant in the dark circle of branches, from the great corral of evergreens in which the public rites of the last night are performed. It is known to the white men who live among these Indians as the hoshkawn dance, from one of the public dances of the last night, in which the Indian jugglers pretend to grow and develop the *hackàn*, or *Yucca baccata*. This last night's performance is varied and interesting and all persons, including whites and Indians of other tribes, are permitted to witness it; but previously, for several days, mystic rites are celebrated in the medicine lodge, to the most of which only the initiated are admitted. Dr. Matthews remained ten days in the Indian camp at Niqotlizi, during which time the shamans admitted him into their medicine lodge and allowed him to observe their rites and practices.

His most interesting discovery on this occasion was that of their system of mythic dry paintings, by which they represent various legends or traditions with dry pigments on the sanded floor of the medicine lodge. A full account of the ceremonies and of the myth on which they are based was prepared by Dr. Matthews and appeared in the Fifth Annual Report of this Bureau.

When the ceremony at Niqotlizi was over he proceeded to a locality in Arizona called by the whites The Haystacks, from the peculiar appearance of the rock formations there. At The Haystacks another great ceremony, probably the second in importance of the Navajo rites, was to take place. Here he again encamped with the Indians and remained until the work of the shamans was done.

The ceremonial observances witnessed on this occasion are, collectively, called by the Navajo *Klèdji-qaçàl*, or chant of the night. They are called by the whites the *Yàybichy* dance, from the name of the principal masked character, *Yèbitcai* or *Gebitcai*, the granduncle of the gods. Like the hoshkawn dance, it has several days of secret rites with elaborate symbolic sand pictures and one night of public dances, less varied and interesting than those of the hoshkawn. Dr. Matthews was permitted to witness the whole performance and to take as many notes and sketches as were necessary.

From The Haystacks Dr. Matthews went to the Indian agency at Fort Defiance, Arizona, where he secured the services of one of the oldest and most learned (in their own peculiar lore) of the Navajo priests, and from him he obtained full explanations of all these rites and of the symbolism of the pictures and masked characters, with a complete recital of the long and elaborate myths on which the ceremonies depend, and the texts and translations of the very numerous songs which form the ritual of the ceremonies.

### **WORK OF DR. H. C. YARROW.**

Dr. H. C. Yarrow, acting assistant surgeon U. S. Army, with the assistance of military details and supplies, in addition to the instruction and facilities provided by this Bureau, started, August 8, 1884, on an expedition into the Territory of Utah, with reference mainly to the exploration of burial mounds and the study of mortuary customs.

Near Choke Cherry Spring a burial cave was discovered, containing the skeletons of three persons, which were secured. Other skeletons, with contents of graves, were obtained near Willow creek; also, an interesting specimen of tree burial.

At Deep creek an explanation of the curious form of water burial was gained from a chief of the Gosiats, to the effect that the bodies of the turbulent and disorderly men of the tribe were thus disposed of to prevent the spirits of these objectionable persons from joining the rest of the tribe after death. Their bodies were sunk in springs and marshy places and kept down by sticks and stones, so that their spirits could never get out.

In the neighborhood of Fillmore a mound was excavated which afforded an admirable example of the beforementioned conversion of a dwelling into a sepulcher. The probability is that the deceased died in his house, which was made of adobe bricks, and that it was at once abandoned and the body left therein, the roof being first removed. The corpse was placed on the floor and covered with a paste of moist clay, on which were placed the mortuary gifts of weapons, utensils, and food. Cottonwood branches were then piled above and set on fire, thus baking the clay crust and charring the several objects. The whole structure had been covered, so that on first examination

the hard surface of burnt clay, 18 inches below the loose earth, appeared to be the floor of a former dwelling.

In the whole of the expedition, which continued into the last days of September, much difficulty was experienced from the suspicion and consequent hostility of the Indians of the localities visited.

### **WORK OF DR. W. J. HOFFMAN.**

Dr. W. J. Hoffman proceeded early in August to Victoria, B. C., where numerous sketches of Haida totem posts and carvings were obtained, in connection with the myths which they illustrated. At this locality attention was paid to the burial customs and osteologic remains of the nearly extinct tribe of Songish Indians.

At Port Townsend sketches were obtained of Thlinkit ivory and wood carvings, clearly indicating the adoption by that tribe of Haida art designs. Here, too, many Indians of British-American tribes were met on their way south to work in the Puyallup hop fields, notable among which was a large number of Haida, whose persons were examined for the purpose of copying the numerous and varied tattoo designs with which they were profusely decorated. Interpretations of many of these characters were obtained from the persons bearing them, as well as from the chief artist of the tribe, together with concise descriptions of the methods and customs in connection with tattooing and the materials used. Drawings were made of a collection of Eskimo pictographs and ivory carvings at the museum of the Alaska Commercial Company and the California Academy of Sciences, San Francisco, Cal.

At Santa Barbara, Cal., Dr. Hoffman discovered some painted pictographs and examined a number which have not yet been published. In several private collections at this place were found interesting relics of the Indians formerly inhabiting Santa Cruz island, the most important of which was a steatite cup containing earthy coloring matter and pricking instruments of bone, which had evidently been used in tattooing. Painted pictographs were also visited in the Azuza cañon, twenty-five miles northeast of Los Angeles.

At Tule Indian Agency, in the deep valleys on the western slope of the Sierra Nevada, sketches of pictographs were made in continuation of work accomplished there two years before. Vocabularies were also obtained from the Waitchumni Indians here located, as well as from the few remaining Santa Barbara Indians at Cathedral Oaks, Santa Barbara county, Cal. By far the greatest amount of pictographic material was collected in Owen's valley, California, where series of petroglyphs are scattered over an arid, sandy desert, the extremes of which are more than twenty miles apart.

## OFFICE WORK.

The work upon a synonymy of the Indian tribes of North America, which has been mentioned to some extent in former reports, has been continued with increased energy.

Every tribe of Indians of any size and importance has been treated of by historians under a variety of names. The sources of these different appellations are manifold. In very many instances the names of tribes or other bodies of Indians communicated by themselves have been imperfectly understood and erroneously recorded; misspelled names and typographical errors have been perpetuated.

Traders, priests, and colonists have called the same tribes by different names and the historian has often added to the confusion by handing down these synonyms as the names of other and different tribes. Not a few tribes well known under established names have received new names upon a change of residence, especially when they have removed to a great distance or have coalesced or allied with other tribes. Added to these and to other sources of confusion are the loose and dissimilar applications of the terms clan, band, tribe, confederacy, and league, the same term having been used with various meanings by different authors.

As a consequence the student of Indian languages and customs finds himself in a tangle, as regards tribal names, which it is beyond the power of the individual worker, unaided, to unravel. The scope of the work in question includes the attempt to trace the several names back to their sources and to ascertain their original and proper application, to define their meaning when possible, and to relegate each tribe under its proper title to the linguistic family to which it belongs. In the completion of this work the whole force of the Bureau assists.

The need of a volume giving the results mentioned has long been felt, and it is believed that it will prove to be one of the most important contributions to the accurate study of Indian history ever made. The classification of the languages of the North American Indians is closely connected with the synonymy of tribal names, each work assisting the other. During recent years the number of students who have directed their attention more or less exclusively to the study of Indian languages has been constantly augmented, and as a result of their labors the number of vocabularies has been correspondingly increased; hence the demand for a more comprehensive and satisfactory classification than now exists.

Prior to Gallatin's time little or nothing had been done in the direction of a systematic classification of Indian languages. In 1836 Gallatin issued his treatise in which he classified all the languages which he was able to study by a direct comparison of vocabularies. His classification was an immense advance over anything previously done and has proved a boon for scholars, having served, indeed, practically as the basis for most of the work in the same line performed since his time. No fixed rules of nomenclature, however, have ever been adopted by linguistic writers, and authors have named and renamed linguistic groups without regard to the names imposed upon the same or similar groups by earlier writers. As a result great confusion has followed not only respecting the status of the various linguistic families, but also respecting the identity of the languages which have served as a basis for the several groups proposed. The remedy for this state of affairs is the adoption, with strict adherence thereto, of a code of nomenclatural rules similar in scope to those prevailing among zoölogists.

There would appear to be no good reason why the rule of priority of name, for instance, should not be followed in linguistic as well as in zoölogic classification, or why the same beneficial result of fixity of nomenclature should not be expected to result from the adoption of this rule in the one case as in the other. Students who may attempt to unravel the many perplexing nomenclatural problems arising from unnecessary change of names will certainly agree that such a rule is no less desirable in linguistics than in zoölogy.

Accordingly, the rule of priority of name, within certain limitations, together with some other rules, has been adopted by the Bureau. These limitations and rules, together with a discussion of the subject, which would

still be premature, may be presented by the Director in his next annual report.

Mr. H. W. HENSHAW, when not in the field, was specially engaged in the organization and details of the office work upon tribal synonymy and linguistic classification above described. A careful examination of all the literature pertaining to these correlated subjects was necessary and also the preparation of tentative tables of synonymy. He has prepared such tables and made in connection with them a brief historical résumé of the literature. Much longer time and the work of the whole official force will, however, be needed for the completion for publication of the results of this vast and complicated undertaking.

Mrs. ERMINNIE A. SMITH was occupied, while not engaged in the field as reported above, in the revision for publication of her Tuscarora dictionary, the material for which had been collected during several years.

Col. GARRICK MALLERY continued the collection and classification of material on the two correlated subjects of sign language and pictographs. His two preliminary papers on those subjects have appeared in former annual reports. It is intended, while increasing the data obtained from the Indian tribes of North America, bearing upon these subjects, to supplement and illustrate the mass of information collected from those tribes by comparison with everything of a similar character to be found in other parts of the world and to publish the results of the collection and study in the form of monographs.

Dr. W. J. Hoffman, when not in the field, continued to assist in the work mentioned.

Mr. JAMES C. PILLING'S preparation of the Bibliography of North American Languages continued during the year. In October and November he visited several libraries in Boston and Providence, for the purpose of clearing up a number of doubtful points. During the year pages 839-1135 were received from the printer, which completed the volume. In the spring a limited number of copies were struck off by the Public Printer, and these have been sent to various libraries, public institutions, and to individuals interested in the subject, for the purpose of obtaining additions and corrections, with the intention, if these should prove to be numerous, of resetting the matter.

Mr. FRANK H. CUSHING was stationed at Washington at the commencement of the fiscal year and was engaged in the classification of his field material in preparation for its publication. During the fall he completed a short paper on Zuñi culture growth as evidenced by studies of Pueblo ceramics, which was published in the Fourth Annual Report of the Bureau. In this paper he maintains, with a large amount of linguistic evidence, that the Zuñi culture is mainly autochthonous, and that its growth, especially the growth of architectural, agricultural, ceramic, and other arts and industries pertaining to it, has been largely accomplished within the desert areas of America which still form the habitat of the Pueblo Indians, and probably, also, within a period more limited than has usually been supposed essential to such development.

He prepared also a paper on the "Ancient province of Cibola and the seven lost cities," in which he not only identifies the seven cities of Cibola above referred to with seven ruins near the present Zuñi village, but also furnishes interesting examples of the permanence of Indian tradition and of its value, when properly used, as a factor, in ethnographic and historic research.

Among the later and perhaps more important results of his studies during the year are investigations of the myths and folk tales abundantly recorded by him during previous years among the Zuñi.

By the extended comparison which he is able to make between these folk tales and myths, now first brought together as a whole, and by the application to their study of the linguistic method employed by him in the preparation of the two papers already mentioned, he is able to trace the growth of mere ideas or of primitive conceptions of natural or biotic phenomena and of physical or animal function into the personæ and incidents which go to make up myths, as well as to trace the influence of these growths on the worship of the Zuñi.

Early in 1885 Mr. Cushing furnished the Director with a schedule of his manuscript, notes, and sketches, and from an examination of this it was deemed advisable that he should continue putting his linguistic material into permanent shape, in order that it might be used as a check on ensuing studies of the sociology and mythology of the Zuñi, as well as for its suggestive value towards the explanation of obscure passages in those

departments of study. This work had progressed but little, however, when a severe illness necessitated its temporary abandonment.

Prof. CYRUS THOMAS, in addition to his administrative duties in charge of the division of mound exploration, was engaged in preparing for publication the results of the operations of that division. The constant arrangement, comparison, and study of the material objects and facts ascertained required his close application. He also commenced the paper presented by him in this volume.

Mr. VICTOR MINDELEFF, in the first part of the fiscal year, completed models of the seven villages of the ancient Province of Tusayan, together with a relief model illustrating the topographical character of the province. The model of Walpi, of this series, was carried out in such a manner as to show on a large scale the character of the rocky mesa on which the town is built. Several types of cliff ruins were also modeled for this series, among them the White House ruin of Cañon de Chelly and the mummy cave of Cañon de la Muerte. After August 1 this work was carried on under the supervision of Mr. Cosmos Mindeff, who also prepared a model of the great Etowah mound from the data of Mr. V. Mindeff's survey; he also furnished several other examples of mounds, with sections, under the direction of Prof. Cyrus Thomas. This work was carried on without interruption until December 7, when Mr. Cosmos Mindeff was ordered to New Orleans, to take charge of the combined exhibits of the U. S. Geological Survey and the Bureau of Ethnology, and was instructed to look after the proper installation of the same in the Government building. He returned to Washington about February 1. During the ensuing four months the small force in the modeling room was engaged in making models of the ancient pueblos of the Chaco, from the plans secured during the preceding summer, as referred to in the report of field work. This work continued until early June, when Mr. C. Mindeff was again ordered to New Orleans to take charge of the packing and shipment of the exhibits of the Geological Survey and Bureau of Ethnology for their return to Washington and for the installation of a portion of the material at the Louisville Exposition. During the interval from February 1 to June 15 Mr. Victor Mindeff was engaged in the preparation of a report on the architecture of the ancient provinces of Cibola and Tusayan, together with the plans and diagrams necessary for its illustration.

This study was based on the large amount of data that had been secured during former field seasons for modeling purposes.

Rev. J. OWEN DORSEY, when not in the field, made nearly 10,000 entries for the Čegiha-English dictionary, and prepared Ponka and Omaha native texts, with free and interlinear translations, in addition to those found in part 1 of vol. 6, Contributions to North American Ethnology. After December 1, 1884, he collated the following vocabularies obtained by him in Oregon, viz: Takelma, Shasti, Applegate Creek, Chastā Costa, Galice Creek, Mulluk, Siuslaw, Lower Umpqua, Yaquina, Klikitat, and one on Smith River, California. He also prepared a list of the villages obtained from the tribes at the Siletz Agency, Oregon.

Mr. ALBERT S. GATSCHET was engaged at the beginning of the fiscal year in revising and perfecting his grammar of the Klamath language of southern Oregon. The phonology was completed and stereotyped, extending from page 200 to 245. He was engaged in correcting proofs of the subsequent section on morphology when he proceeded to the Southwest, as elsewhere reported, to investigate several languages spoken there, the affinities of which had not before been ascertained.

Mr. W. H. HOLMES, as in previous years, has supervised the illustrations of the Bureau publications. He also continued his archæologic studies, chiefly in the department of ceramics, the character of which is shown by his papers in this volume. He was in charge of the preparation of exhibits for the expositions at New Orleans, Louisville, and Cincinnati; but, owing to the pressure of other duties, much of this work was intrusted to Mr. Cosmos Mindeleff, who was assisted materially by Mr. Victor Mindeleff. The most important feature of the exhibits consisted of models of plaster and papier mâché of the pueblo towns and cliff houses of New Mexico and Arizona.

Aside from the models, exhibits of ethnologic and archæologic materials were made. A large and important collection of objects of pueblo art was obtained by Mr. James Stevenson, but much of it failed to reach Washington in time for exhibition purposes, and a series of similar objects, already classified and labeled, was selected from the National Museum and forwarded to New Orleans. A valuable collection of the ancient fictile products of Tusayan belonging to Mr. Thomas Keam was also utilized in perfecting the exhibits of Pueblo art.

Archæologic materials from other sections of the country were placed on exhibition, notably a superb collection of prehistoric relics from the province of Chiriqui, Panama, which was purchased for the purpose.

The collections of ethnologic and archæologic material made during the year are of unusual importance and magnitude. This is chiefly due to the facilities afforded by the New Orleans Exposition fund, a liberal portion of which was devoted to the collection and purchase of objects of permanent value to the Government and to science. The collections made by Mr. Stevenson in Zuñi and Acoma comprise upward of four thousand pieces, chiefly objects of clay, but including other classes of products. The collection of prehistoric relics obtained by Mr. J. A. McNiel from the tombs of Chiriqui is one of the most important and complete series of ancient American products to be found in any country, and must prove of great value to students.

Mr. Victor Mindeleff secured a small series of relics from the ancient ruins of northern New Mexico and Arizona, and Dr. H. C. Yarrow added some objects of archæologic and ethnologic interest from central Utah. Mrs. Erminnie A. Smith procured a number of articles of shell, illustrating the modern manufacture of wampum in New Jersey; a small collection of fragmentary pottery from the eastern shore of Maryland was presented by Mr. Joseph D. McGuire, of Ellicott City; and Mr. Holmes secured a series of articles, including arrowheads, shell implements, and pottery, from the island of Nantucket. Mound explorations, conducted by Dr. Cyrus Thomas, yielded a valuable series of objects of stone and clay. An unusually interesting series of the earthen vessels of the ancient pueblo races was secured by Mr. E. W. Nelson in eastern central Arizona. The greater part of the abovementioned material has already been catalogued and turned over to the U. S. National Museum.

Dr. H. C. YARROW, acting assistant surgeon, U. S. Army, besides his field explorations described, continued to collect information relative to the mortuary customs of North American Indians. Of the material gathered, a considerable portion has been forwarded by various persons throughout the country in answer to the circular sent out early in the last year, but much has also been derived from the published works on anthropologic subjects, including scientific journals and reports. Numerous authorities have been

consulted and much time has been devoted to the consideration of the many theories advanced to account for certain peculiar rites and customs.

Mr. CHARLES C. ROYCE continued during the year the preparation of a historical atlas of Indian cessions. The boundaries of the various cessions of land by the different Indian tribes were traced out and located upon the maps of the States and Territories left uncompleted at the date of the last annual report. All that remains to be done in completing the atlas for publication is to transcribe, with considerable elaboration, the historical and descriptive notes pertaining to the various cessions, and to make, from the rough working sheets, legible copies of the maps showing the boundaries of the cessions within the States of California, Oregon, Nevada, and Texas and the Territories of Washington, Idaho, Montana, Wyoming, Utah, New Mexico, Arizona, and Dakota. Most of these States and Territories will each require two maps, showing respectively the primary and secondary cessions. The work will be finished as rapidly as possible.

## **ACCOMPANYING PAPERS.**

The present volume contains papers the subject matter of which may be classified under the grand divisions of Technology, Philosophy, Sociology, and Ethnography.

They are all prepared by experts of recognized authority in their several lines of research and are illustrated to the degree required by the text for full understanding, the number of figures presented being 548, besides ten full page plates. Special mention of each of these papers follows in their order as printed.

**ANCIENT ART OF THE PROVINCE OF CHIRIQUI,  
COLOMBIA, BY  
WILLIAM H. HOLMES.**

The archæology of Chiriqui should be studied, not only for comparison with that of the territory comprised in the present political divisions of North America, but because geographically the province should be considered as a part of the North American continent. Until recently this isthmian region was little known, the explorations for railroads and canals having furnished the first valuable accounts of its modern inhabitants and the relics left by former occupants.

The National Museum now contains a large and precious collection of archæologic material from the province, chiefly obtained by Mr. J. A. McNiel during years of enthusiastic labor. The information derived and the lessons to be learned from this collection, together with all particulars relating thereto gathered from other sources, are now presented in this paper by Mr. W. H. Holmes. His work in the classification of the immense number of objects and in the elucidation of their functions, material, construction, forms, and decorations has been careful and comprehensive. His manifest success has been owing to his artistic insight and skill as well as to his archæologic training. His ability in both fields can be appreciated by an examination of the 287 illustrations in his paper, considered not only as to their number, but as to their instructive arrangement in his text.

The objects of ancient art found in Chiriqui are, as elsewhere in North America, derived almost entirely from graves. The cemeteries, apart from their contents and the mode of sepulture, constitute in themselves topics of interest which are discussed and illustrated in the paper. Another curious feature is that the objects buried generally appear to have been manufactured for mortuary purposes and not for use by the living. A general review of the contents of the graves shows that the ancient inhabitants were skillful in the manipulation of stone, gold, copper, and clay, and tombs of undoubtedly great antiquity yield evidence of long continued culture.

It also appears that, while the art of the old peoples of the isthmus can in some respects be connected with that of adjacent regions in North America, in others it is remarkable for individuality. Ornaments of stone were seldom used by them and those of gold and copper were common. The articles of gold which the graves have yielded in large quantities to explorers during the last quarter of a century, and for which only they have until recently been searched, have generally been considered to be mere ornaments, but they probably had a fetichistic origin.

It is remarkable that no weapon, tool, or utensil of metal has been noticed. The objects were generally formed by casting in molds, which was done with considerable skill, and gilding, or at least plating, was practiced. The art of alloying also appears to have been understood.

The use of metals does not appear early in the order of technology, and an advanced degree of culture is generally attained before the casting of any metal is attempted. Without allowing too much weight to any argument based upon the surprising skill of these people in plating and alloying, the evidence of technical skill in general, together with the conceptions embodied in their art, proves conclusively that it was the product of a long period of experiment and progress.

The pottery of Chiriqui is to be noted for the perfection of its technique, its high specialization of form, and its conventional use of a wide range of decorative motives. Its forms present many striking analogies to the wheel made ware of the Mediterranean, regarded as classic.

The mythologic stage of the builders of these graves is shown by the fact that in their ceramic art there is no attempt to render the human face or figure with accuracy. The personages of their religious philosophy were zoöomorphic and some of their forms may be discerned by a skillful analyst in or on all the ornaments and vessels. On each of the latter all decorative devices and delineations have some reference to the mythic creature associated with the vessel and its functions.

Mr. Holmes has made an important discovery in the evolution of decoration in Chiriqui from which are deduced instructive generalizations of wide application. All the decorations originate (doubtless under the influence of the stage reached in mythologic philosophy) in life forms of animals, none being vegetal. Coming from mythologic concepts they are significant and

ideographic, and coming from nature they are primarily imitative and non-geometric. Nevertheless the agencies of modification inherent in the practice of art through its mechanical conditions are such that the animal forms early employed have changed into conventional decorative devices, among which are the meander, scroll, fret, chevron, and guilloche.

That this was the course of evolution of the classic forms of ornaments is not asserted; indeed, it is not necessary to form such a hypothesis, as by the interacting principles, well classified by Mr. Holmes, the course by which the same result was accomplished may have been wholly diverse. It is, however, shown that this was in all probability the particular and independent course in one region of America, being in that respect in distinct contrast to other art regions, such as that of the Pueblos, where the rise of geometric figures through technologic channels is equally obvious. It follows that in seeking to divide peoples by the criteria of their decorative arts the examination must embrace what is far more fundamental than a mere comparison of their finished products: these may be and are markedly similar without any evidence of transmission, and when in fact by deeper study the ascertained separate courses of development preclude such transmission.

# **A STUDY OF THE TEXTILE ART IN ITS RELATION TO THE DEVELOPMENT OF FORM AND ORNAMENT, BY W. H. HOLMES.**

For several years Mr. Holmes has been engaged in the study of the ancient and existing art of the North American Indians, and has published in the annual reports of this Bureau a number of elaborate essays upon the art of specified peoples and regions.

In the present paper he submits the comprehensive results of his studies in one great branch, the textile art, and treats chiefly of its esthetic relations as distinct from those of construction and function, so far as they can be separately discussed.

He has been fortunate in the character of the material studied. In America there is yet found a great body of primitive, indigenous, and independent art, almost uncontaminated by the complex phenomena, processes, and conditions which elsewhere obscure its origin and development. To a knowledge of American art acquired by long study Mr. Holmes adds a mental equipment exceptionally qualifying him for its philosophic discussion. His conclusions therefore, presented with ample evidence and explained by illustrations, are to be received as those of a recognized authority, although they may disturb some sentimental and metaphysical fancies concerning abstract beauty in form, color, and design.

It is not contended that the earliest concepts of beauty originated with textile art. On the contrary, it is probable that the first esthetic attempts were in the line of personal decoration, such as paints on the skin and pendants and feathers disposed about the person. But as the textile art appears early and widely in culture it is believed that the association of esthetic concepts with it very generally preceded their association with other arts. Having thus the start in the field, its nature was full of suggestions of embellishment, while it was fixed in its method of expression. The technique therefore shaped and directed the esthetic concept and became the parent of much geometric ornament.

Mr. Holmes gives an instructive analysis of the forces and influences inherent in the textile art, the first lessons of which are order, uniformity, and symmetry; he shows how the necessities of technique determine ideas of the beautiful in linear geometric forms and how taste in selecting certain ornaments as the most beautiful is simply choosing that product which in the evolution of art gave it character and power.

The influence of textile ornament upon other forms of art, such as architecture and sculpture, is discussed, as also the manner in which extrinsic decorative elements are remodeled in accordance with the rules of textile combination. The paper, however, does not undertake to cover the whole field of the development of form and ornament, being confined to the relation of the textile art thereto, and similar studies in all other grand divisions of art must be made before the relative importance of all their forces and tendencies can be estimated. But the laws of evolution in all art closely correspond, and the present paper is eminently instructive to all students of the esthetic.

## **AIDS TO THE STUDY OF THE MAYA CODICES, BY CYRUS THOMAS.**

That Prof. Cyrus Thomas has long been engaged in the examination of the few Maya records in existence is known from his former works, "A study of the Manuscript Troano" and "Notes on certain Maya and Mexican manuscripts," both published by the Bureau of Ethnology.

The object of the present paper is to give information of some original discoveries and to present some explanations not brought forward by Professor Thomas in his former papers.

The records of Maya and Mexico yet challenge students with unsolved problems similar to those which in the writings of Egypt and Assyria have perplexed so many generations. The translation of the paleographic literature of this continent may be expected to throw light on the past of America, in some degree reproducing the brilliant result which has attended the translation of the hieroglyphs of the eastern hemisphere. Long and laborious comparisons, together with the trial of successive hypotheses, will be necessary to the decipherment of our aboriginal manuscripts, and but few competent persons are actively engaged in the work. It becomes, therefore, the duty of any one whose discoveries tend to clear up even minor points of the great problem to furnish them to his fellow laborers, and thereby limit the remaining field of investigation. In this paper Professor Thomas supplements his former work.

## **OSAGE TRADITIONS, BY REV. J. OWEN DORSEY.**

This paper contains an account of a secret society of seven degrees, still existing among the Osage, in which the traditions of the people have been preserved. The author, by his skill and personal influence, has obtained and now furnishes two of these traditions in the original language, with an interlinear and a free translation of each and with explanatory remarks.

The traditions are both cosmologic and sociologic, and are admirable examples of Indian philosophy. The existence of secret associations, periodically celebrating religious mysteries, and of shamanistic orders, which, by ceremonies, pictographs, and chants, have preserved in more or less purity the traditions of their ancestors, has been vaguely known for some years, but until lately no accurate or indeed intelligent account of them has been secured.

The exertions of several of the officers of this Bureau have been successful in obtaining full details and clear explanations both of the traditions and the ceremonials of several of the Indian tribes, notably those of the Zuñi and the Navajo, published in former annual reports. The present paper by Mr. Dorsey takes an important place in this new collection of materials for the study of Indian philosophy, from which valuable results have been already acquired.

## **THE CENTRAL ESKIMO, BY DR. FRANZ BOAS.**

For the express purpose of personal exploration and examination, the author of this important paper spent a considerable time in the region of which he treats. His course of travel was to Cumberland sound and Davis strait. The grand division of the Eskimauan linguistic family, inhabiting nearly the whole range of the Arctic-American coast, which has been classed as Central Eskimo, occupies the northeastern part of the continent and the eastern islands of the Arctic-American archipelago. It inhabits, at Smith sound, the most northern countries in which man has been known to dwell. Its southern and western boundaries are about Fort Churchill, the middle part of Back river, and the coast west of Adelaide peninsula.

Dr. Boas gives an admirable account of the topography of the region and of the distribution, tribal divisions, and numbers of the inhabitants. His work is replete with valuable statements in minute detail and with acute suggestions regarding their habits and customs. Their peculiar and ingenious weapons, implements, and utensils are fully described and illustrated. His account of their religious practices and beliefs, supplemented by translations of their myths and legends, is equally entertaining and instructive.

In connection with his observations made through original research, Dr. Boas presents the result of a close study and analysis of the work of former explorers in this field, by which his contribution to the study of this interesting hyperborean people will command additional attention.

## FINANCIAL STATEMENT.

*Table showing amounts appropriated and expended for North American ethnology for the fiscal year ending June 30, 1885.*

| Expenses.                                                             | Amounts<br>expended. | Amount<br>appropriated. |
|-----------------------------------------------------------------------|----------------------|-------------------------|
| A. Services                                                           | \$30,433.55          |                         |
| B. Traveling expenses                                                 | 3,716.14             |                         |
| C. Transportation of property                                         | 354.12               |                         |
| D. Field subsistence                                                  | 198.42               |                         |
| E. Field supplies and expenses                                        | 535.45               |                         |
| F. Field material                                                     | 197.71               |                         |
| G. Instruments                                                        | 49.25                |                         |
| H. Modeling material                                                  | 40.11                |                         |
| I. Photographic material                                              | 306.71               |                         |
| K. Books and maps                                                     | 355.85               |                         |
| L. Stationery and drawing material                                    | 15.70                |                         |
| M. Illustrations for reports                                          | 668.64               |                         |
| N. Articles for distribution to Indians                               | 23.69                |                         |
| O. Office furniture                                                   | 59.67                |                         |
| P. Office supplies and repairs                                        | 36.61                |                         |
| Q. Manuscripts                                                        | 395.00               |                         |
| R. Correspondence                                                     | 15.43                |                         |
| S. Specimens                                                          | 71.00                |                         |
| T. Collection of material for<br>classification of the Indians in the | 1,326.61             |                         |

|                                                                     |           |             |
|---------------------------------------------------------------------|-----------|-------------|
| United States<br>Balance on hand to meet<br>outstanding liabilities | 1,200.34  |             |
|                                                                     | 40,000.00 | \$40,000.00 |

## **ACCOMPANYING PAPERS.**

# INDEX.

## A.

|                                                      | <b>Page.</b>                |
|------------------------------------------------------|-----------------------------|
| Acoma, collections of J. Stevenson from              | <u><a href="#">XLIX</a></u> |
| Adelung, J. C., cited                                | 262                         |
| Adlet and Qadlunait, origin of the                   | 637                         |
| Adlet or Erqigdlit                                   | 640                         |
| Aggomiut Eskimo tribe, situation and subdivisions of | 442-444                     |
| Aglio, Augustina, fac simile of Dresden Codex by     | 263-266                     |
| Agutit Eskimo tribe, situation of                    | 450, 451                    |
| Aivillirmiut Eskimo tribe, situation of              | 445-450                     |
| Akudnirmiut Eskimo tribe, situation of               | 440-442                     |
| Akuliarmiut Eskimo tribe, situation of               | 421                         |
| Alaskan Indians, illustration of ornamentation by    | 199                         |

|                                                                      |                                           |
|----------------------------------------------------------------------|-------------------------------------------|
| Alligator, utilization of, in Chiriquian art                         | 130-140, 166,<br>173-176, 178,<br>80, 183 |
| American Museum of Natural History, acknowledgments<br>to            | 409                                       |
| [American Museum of Natural History], figured<br>specimens from      | 472, 517                                  |
| Ancient art of the province of Chiriqui, paper by W. H.<br>Holmes on | 13-187                                    |
| Ancon, Peru, examples of ornamentation from graves at                | 212, 230, 231,<br>236, 243, 248           |
| Anderson and Stewart, cited                                          | 458, 459                                  |
| Apache, illustrations of ornamentation by                            | 198, 213, 223                             |
| Ardnainiq, fabulous tribe in Eskimo tradition                        | 640                                       |
| Arrowpoints and spearheads of Chiriqui                               | 34                                        |

## **B.**

|                                                |          |
|------------------------------------------------|----------|
| Back, cited                                    | 485      |
| Baffin Land, description of                    | 415, 416 |
| [Baffin Land], distribution of tribes in       | 421-444  |
| [Baffin Land], traditions of, with comparisons | 641-643  |

|                                                                                                   |                                                       |
|---------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Balboa, ornaments captured by                                                                     | 35                                                    |
| Berliner Gesellschaft für Anthropologie etc.,<br>Verhandlungen der, cited                         | 409, 616                                              |
| Bessels, Emil, cited                                                                              | 412, 460, 486                                         |
| Bibliography of the Languages of the North American<br>Indians                                    | <a href="#">XXIV</a> -XXVI                            |
| Bickford, F. T., field work of                                                                    | <a href="#">XXVIII</a>                                |
| Bill Nix (W. P. Matthews), Osage traditions dictated by                                           | 377                                                   |
| Black incised group of Chiriquian pottery                                                         | 80                                                    |
| Boas, F., remarks on paper respecting Central Eskimo by<br>[Boas, F.], paper on Central Eskimo by | <a href="#">LVI</a> , <a href="#">LVII</a><br>399-669 |
| Bollaert, W., cited                                                                               | 41, 45                                                |
| Boothia Felix and Back River, tribes of                                                           | 452-459                                               |
| Böttiger, C. A., mention of Dresden Codex by                                                      | 262                                                   |
| [Böttiger, C. A.], controversy with Abert concerning<br>Dresden Codex                             | 267                                                   |
| Brasseur, copy of the Manuscript Troano by<br>[Brasseur], cited                                   | 284, 286, 343<br>350                                  |
| British Guiana Indians, illustrations of ornamentation by                                         | 217                                                   |

## C.

|                                                                                                       |                                                       |
|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| Calendar system, tabular view of                                                                      | 270-274                                               |
| Castillo del Oro, name given by Columbus to Chiriqui                                                  | 35                                                    |
| Øegiha-English dictionary                                                                             | <a href="#">XLVIII</a>                                |
| Celts, collection of, from Chiriqui                                                                   | 29-34                                                 |
| Central Eskimo, remarks on paper of F. Boas on<br>[Central Eskimo], paper on, by F. Boas              | <a href="#">LVI</a> , <a href="#">LVII</a><br>399-669 |
| Charency, H. de, cited                                                                                | 282                                                   |
| Chimu, Peru, ornamentation of "hall of arabesques" at                                                 | 251, 252                                              |
| Chiriqui, collections from tombs at<br>[Chiriqui], ancient art of the province of, by W. H.<br>Holmes | <a href="#">XLIX</a><br>13-187                        |
| Cibola, identification of ancient cities of                                                           | <a href="#">XLVI</a>                                  |
| Clallam Indians, illustrations of ornamentation by                                                    | 207                                                   |
| Codex Cortesianus, similarity of, to Manuscript Troano<br>and Dresden Codex                           | 286                                                   |
| Collinson, cited                                                                                      | 503                                                   |

|                                                           |                                                                   |
|-----------------------------------------------------------|-------------------------------------------------------------------|
| Color in textile art                                      | 201, 202                                                          |
| Color phenomena in textile ornament                       | 215-232                                                           |
| Comanche Indians, linguistic work of A. S. Gatschet among | <a href="#"><u>XXXIV</u></a>                                      |
| Costa Rica, origin of name of                             | 35                                                                |
| Cranz, D., cited                                          | 412, 586, 590                                                     |
| Cumberland Sound, description of settlements of           | 428-440                                                           |
| Curtin, J., linguistic field work of                      | <a href="#"><u>XXXVII</u></a> ,<br><a href="#"><u>XXXVIII</u></a> |
| Cushing, F. H., office work of                            | <a href="#"><u>XLVI</u></a>                                       |

## D.

|                                                                     |                 |
|---------------------------------------------------------------------|-----------------|
| Dakota, organization of the                                         | 396             |
| Darien, capture of, by Balboa                                       | 35              |
| Davis Strait Indian tribes, snow houses of                          | 541-544         |
| Dease and Simpson, cited                                            | 458             |
| De Zeltner, A. See Zeltner, A. de. Diller, J. S., acknowledgment to | 21, <i>note</i> |

Dogs and sledges of Eskimo 529-538

Dorsey, J. O., linguistic field work of [XXXVI](#)

[Dorsey, J. O.], office work of [XLVIII](#)

[Dorsey, J. O.], remarks on paper respecting Osage traditions by [LV](#), [LVI](#)

[Dorsey, J. O.], paper by, on Osage tradition 373-397

Dresden Codex, numerals in 261-338

Drums of ancient Chiriqui 157, 160

## E.

Ebert, F. A., description of Dresden Codex by 263

[Ebert, F. A.], controversy with Böttiger concerning Dresden Codex 267

Eenoolooapik, cited 410, 425, 464

Egede, H., cited 412

El Dorado, origin of 35

Ellesmere Land, natives of 459, 460

Emigration of the Sagdlirmiut 616-620

Emmert, J. W., field work of [XXVI](#), [XXVII](#)

Erdmann, F., cited 412, 597

Eskimo, the Central, by F. Boas 399-669

## F.

Falkenstein, K. C., preservation of Dresden Codex by 268

Field work [XXVI](#)-XLII

Figurines of Chiriquian art 151-153

Fillmore, exhumation of sepulcher at [XLI](#)

Financial statement [LVIII](#)

Fishing, Eskimo methods, of 513-516

Fleischer, H. L., mention of Dresden Codex by 263

Flight to the moon 598, 599

Form in textile art and its relation to ornament, with  
illustrations from Indian work 196-201

Förstemann, E., citation from Die Mayahandschrift of 261-269  
[Förstemann, E.], cited 272, 278, 280,  
281, 283, 290,  
292, 293, 300,  
301, 302, 303,  
304, 305, 320,  
322, 329, 330,  
339, 340

|                                  |                      |
|----------------------------------|----------------------|
| Fowke, G., field work of         | <a href="#">XXVI</a> |
| Frobisher, M., cited             | 410, 469, 558        |
| Frobisher Bay, use of, by Eskimo | 423                  |

## G.

|                                                                   |                                              |
|-------------------------------------------------------------------|----------------------------------------------|
| Gallatin, A., mention of classification of Indian languages<br>by | <a href="#">XLIV</a>                         |
| Garlic, C. A., field work of                                      | <a href="#">XXIX</a>                         |
| Gatschet, A. S., linguistic field work of                         | <a href="#">XXXIII-</a><br>XXXVI             |
| [Gatschet, A. S.], office work of                                 | <a href="#">XLVIII</a>                       |
| Geography, Eskimo knowledge of                                    | 643-647                                      |
| Geometric design, relations of, to textile ornament               | 202-244                                      |
| Gilder, W. H., cited                                              | 411, 456, 457,<br>458, 459, 466,<br>498, 522 |
| Glossary of Eskimo terms                                          | 663-669                                      |
| Gordon, A. R., cited                                              | 412, 463                                     |
| Gosiats, water burial among                                       | <a href="#">XLI</a>                          |

|                                                |          |
|------------------------------------------------|----------|
| Götze, J. C., preservation of Dresden Codex by | 261      |
| [Götze, J. C.], biographical sketch of         | 261, 262 |

## H.

|                                               |                                                                                                                                                                                                                                       |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Haida Indians, art among                      | <a href="#">XLI</a> , <a href="#">XLII</a>                                                                                                                                                                                            |
| Hall, C. F., acknowledgments to               | 409                                                                                                                                                                                                                                   |
| [Hall, C. F.], cited                          | 411, 422, 432,<br>442, 443, 444,<br>445, 446, 447,<br>448, 449, 450,<br>452, 456, 457,<br>459, 462, 463,<br>464, 486, 499,<br>503, 509, 547,<br>578, 583, 589,<br>594, 595, 596,<br>601, 602, 606,<br>607, 608, 611,<br>614, 615, 639 |
| Hallock, W., on Chiriquian methods of casting | 38                                                                                                                                                                                                                                    |
| Handled group of Chiriquian pottery           | 90-97                                                                                                                                                                                                                                 |
| Harpoons of Eskimo, mode of constructing      | 489-494                                                                                                                                                                                                                               |
| Henshaw, H. W., linguistic field work of      | <a href="#">XXXI</a> -<br>XXXIII                                                                                                                                                                                                      |
| [Henshaw, H. W.], office work of              | <a href="#">XLV</a>                                                                                                                                                                                                                   |

|                                                                                                                 |                                               |
|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| Herrera, cited                                                                                                  | 35                                            |
| Hewett, J. N. B., field work of                                                                                 | <a href="#">XXX</a> , <a href="#">XXXI</a>    |
| Hoffman, W. J., field work of                                                                                   | <a href="#">XLI</a> , <a href="#">XLII</a>    |
| [Hoffman, W. J.], office work of                                                                                | <a href="#">XLV</a>                           |
| Holmes, W. H., office work of                                                                                   | <a href="#">XLVIII</a> , <a href="#">XLIX</a> |
| [Holmes, W. H.], abstract of paper on ancient art of the<br>province of Chiriqui by                             | <a href="#">LI</a> -LIV                       |
| [Holmes, W. H.], abstract of paper on textile art in its<br>relation to the development of form and ornament by | <a href="#">LIV</a> , <a href="#">LV</a>      |
| [Holmes, W. H.], paper by, on ancient art of the province<br>of Chiriqui                                        | 13-187                                        |
| [Holmes, W. H.], paper by, on textile art in its relation to<br>the development of form and ornament            | 189-252                                       |
| Huacals, exploration of, in Chiriqui                                                                            | 16, 17                                        |
| Hudson Bay, tribes of western shore of                                                                          | 444-452                                       |
| Hudson Bay district, geographic description of                                                                  | 414-418                                       |
| Hudson Bay Indians, snow houses of                                                                              | 547                                           |
| Humboldt, A. von, notice of Dresden Codex by                                                                    | 262, 263                                      |
| Hunting, Eskimo methods of                                                                                      | 471-513                                       |

## I.

|                                                   |                            |
|---------------------------------------------------|----------------------------|
| Igdlumiut Eskimo tribe, situation of              | 463                        |
| Igdlungajung, fabulous tribe in Eskimo tradition  | 640                        |
| Igimarasugdjuqdjuaq the cannibal                  | 633, 634                   |
| Iglulik Eskimo tribe, snow houses of              | 546, 547                   |
| Iglulirmiut Eskimo tribe, situation of            | 444                        |
| Ijirang, fabulous people in Eskimo tradition      | 640                        |
| Indian tribes, synonymy of                        | <a href="#">XLIII</a> -XLV |
| Inuarudligang, fabulous tribe in Eskimo tradition | 640                        |
| Inugpaqduqdjualung                                | 638                        |
| Inuit race, divisions of                          | 420                        |
| Iowa, secret society among                        | 396                        |

Ititaujang; 615-618

|                                      |     |
|--------------------------------------|-----|
| Itivimiut Eskimo tribe, situation of | 463 |
|--------------------------------------|-----|

## J.

|                                           |                      |
|-------------------------------------------|----------------------|
| Jones, Mrs. J. A., Onondaga manuscript of | <a href="#">XXXI</a> |
|-------------------------------------------|----------------------|

## K.

|                                                           |                                                               |
|-----------------------------------------------------------|---------------------------------------------------------------|
| Kadlu the thunderer                                       | 600                                                           |
| Kaiowē language, researches of A. S. Gatschet respecting  | <a href="#"><u>XXXIV</u></a> ,<br><a href="#"><u>XXXV</u></a> |
| Kalopaling                                                | 620, 621                                                      |
| Kangivamiut Eskimo tribe, situation of                    | 463                                                           |
| Kansa, secret society among                               | 396                                                           |
| Kayak, construction of                                    | 486-489                                                       |
| Keam, T., Tusayan products collected by                   | <a href="#"><u>XLIX</u></a>                                   |
| Kingnaitmiut Eskimo tribe, situation of                   | 424                                                           |
| Kingsborough, Lord, Dresden Codex copied by order of      | 262                                                           |
| [Kingsborough, Lord], Mexican Antiquities of, cited       | 266                                                           |
| Kinipetu or Agutit Eskimo tribe, situation of             | 450, 451                                                      |
| Kiviung                                                   | 621                                                           |
| Klamath Indians, illustrations of ornamentation by        | 208, 209, 227                                                 |
| Klamath language, work by A. S. Gatschet on grammar<br>of | <a href="#"><u>XLVIII</u></a>                                 |
| Kleinschmidt, Eskimo orthography of                       | 413                                                           |

|                         |                                                                                                                      |
|-------------------------|----------------------------------------------------------------------------------------------------------------------|
| Klutschak, H. W., cited | 411, 448, 449,<br>451, 457, 458,<br>459, 466, 502,<br>509, 510, 516,<br>552, 553, 570,<br>582, 595, 596,<br>614, 615 |
|-------------------------|----------------------------------------------------------------------------------------------------------------------|

|                                         |     |
|-----------------------------------------|-----|
| Kouksoarmiut Eskimo tribe, situation of | 463 |
|-----------------------------------------|-----|

|                                                         |                                                                                              |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Kumlien, L., acknowledgments to<br>[Kumlien, L.], cited | 409<br>412, 471, 474,<br>475, 482, 483,<br>524, 549, 550,<br>567, 589, 596,<br>606, 607, 610 |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------|

|                                                            |    |
|------------------------------------------------------------|----|
| Kunz, G. F., on use of insects as models in casting metals | 38 |
| [Kunz, G. F.], on Chiriquian methods of plating            | 39 |

## L.

|                                     |                             |
|-------------------------------------|-----------------------------|
| La Fort, D., Onondaga manuscript of | <a href="#"><u>XXXI</u></a> |
|-------------------------------------|-----------------------------|

|              |     |
|--------------|-----|
| Landa, cited | 348 |
|--------------|-----|

|                                    |          |
|------------------------------------|----------|
| Landa's alphabet, insufficiency of | 259, 347 |
|------------------------------------|----------|

|                |     |
|----------------|-----|
| Lepsius, cited | 418 |
|----------------|-----|

|                                                        |                               |
|--------------------------------------------------------|-------------------------------|
| Lipan Indians, linguistic work of A. S. Gatschet among | <a href="#"><u>XXXIII</u></a> |
|--------------------------------------------------------|-------------------------------|

"Lost color" of Chiriquian art, nature of 86

Lost color group of Chiriquian pottery 113-130

Lyon, G. F., cited 410, 451, 463,  
487, 497, 511,  
579, 585, 586,  
587, 588, 589,  
590, 592, 593,  
610, 612, 613,  
614, 615

## **M.**

M'Donald, A., cited 410

McGuire, J. D., collections of [L](#)

Maclean, J. P., field work of [XXVII](#)

M'Clintock, Captain, cited 411, 455, 456,  
458

McCloud River Indians, illustrations of ornamentation by 221

McNiel, J. A., collection of relics from tombs of Chiriqui [XLIX](#)  
by

[McNiel, J. A.], archæologic work of, in Chiriqui 14, 15, 20

[McNiel, J. A.], cited 17, 22, 23, 27,  
31, 40, 41, 43,  
46, 107

|                                                          |                                                   |
|----------------------------------------------------------|---------------------------------------------------|
| Mallery, G., office work of                              | <a href="#">XLV</a>                               |
| Manufactures, Eskimo                                     | 516-526                                           |
| Manuscript Troano, copy of, by Brasseur                  | 285, 286, 343                                     |
| [Manuscript Troano], study of, by C. Thomas, cited       | 339, 343, 344,<br>345, 350, 365,<br>366, 367, 370 |
| Maroon group of Chiriquian pottery                       | 107-109                                           |
| Mason, O. T., acknowledgments to                         | 409                                               |
| Matthews, W., field work of                              | <a href="#">XXXVIII</a> -XL                       |
| Matthews, W. P. (Bill Nix), Osage traditions dictated by | 377                                               |
| Maya and Mexican manuscripts, C. Thomas on, cited        | 280                                               |
| Maya Codices, aids to the study of, by C. Thomas         | 253-371                                           |
| Mealing stones of Chiriqui                               | 25-27                                             |
| Merritt, J. K., cited                                    | 14, 16, 49                                        |
| [Merritt, J. K.], exploration of Bugaba cemetery by      | 17, 18, 20                                        |
| Metates of Chiriqui, nature and use of                   | 25-27                                             |
| Mexican Antiquities, by Lord Kingsborough, cited         | 266, 267                                          |

|                                                             |                                                                                        |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Middleton, J. D., field work of                             | <a href="#">XXVI-</a><br>XXVIII                                                        |
| Mindeleff, C., office work of                               | <a href="#">XLVII,</a><br><a href="#">XLVIII</a>                                       |
| Mindeleff, V., field work of                                | <a href="#">XXIX,</a> <a href="#">XXX</a>                                              |
| [Mindeleff, V.], office work of                             | <a href="#">XLVII,</a><br><a href="#">XLVIII</a>                                       |
| [Mindeleff, V.], collections of                             | <a href="#">XLIX</a>                                                                   |
| Mintzer, W., acknowledgments to                             | 409                                                                                    |
| Modoc Indians, linguistic work among                        | <a href="#">XXXVI,</a><br><a href="#">XXXVII</a>                                       |
| Moki, illustrations of ornamentation by                     | 197, 205, 224,<br>225, 226, 238,<br>240                                                |
| Moravian missionaries, cited                                | 463                                                                                    |
| Mound explorations, field work on                           | <a href="#">XXVI-</a><br>XXVIII                                                        |
| Mountain chant of Navajo Indians                            | <a href="#">XXXIX,</a> <a href="#">XL</a>                                              |
| Museum für Völkerkunde, Berlin, acknowledgments to          | 409                                                                                    |
| [Museum für Völkerkunde, Berlin], figured specimens<br>from | 472, 473, 474,<br>477, 479, 480,<br>481, 483, 486,<br>487, 488, 496,<br>508, 513, 514, |

515, 518, 519,  
520, 523, 531,  
532, 554, 555,  
556, 557, 565,  
566, 567, 568,  
569, 570, 571,  
576, 613, 634,  
644

Music and poetry of the Eskimo 648-658

## N.

Nadaillac, Marquis, cited 14, 38

[Nadaillac, Marquis], on Chiriquian methods of casting 38

Na-ishi Apache Indians, linguistic work among [XXXV](#)

Narwhal, origin of the 625-627

National Museum, acknowledgments to 409

[National Museum], figured specimens from  
474, 479, 480,  
481, 487, 488,  
489, 490, 492,  
493, 494, 495,  
496, 502, 503,  
504, 505, 506,  
507, 512, 513,  
515, 516, 518,  
521, 522, 523,  
524, 525, 526,  
530, 531, 532,  
535, 539, 555,

|                                                                                 |                                            |
|---------------------------------------------------------------------------------|--------------------------------------------|
|                                                                                 | 556, 559, 560,<br>563, 565, 566,<br>576    |
| Navajo Indians, field work of W. Matthews among                                 | <a href="#">XXXVIII</a> -XL                |
| Navigation, Eskimo proficiency in                                               | 643                                        |
| Needlecases (?) of Chiriqui                                                     | 150                                        |
| Nelson, E. W., collection of earthen vessels from eastern<br>central Arizona by | <a href="#">L</a>                          |
| Netchillirmiut Eskimo tribe, situation of                                       | 452-458                                    |
| New Granada, burial customs in                                                  | 19, 20                                     |
| Niqotlizi, ceremonies at                                                        | <a href="#">XXXIX</a> , <a href="#">XL</a> |
| Norris, P. W., field work and death of                                          | <a href="#">XXVI</a>                       |
| Northeastern America, geography of                                              | 414-418                                    |
| North Greenlanders                                                              | 460                                        |
| Northwest Coast Indians, illustrations of ornamentation<br>by                   | 213, 218, 227,<br>230                      |
| Nourse, cited                                                                   | 452                                        |
| Nugumiut Eskimo tribe, situation of                                             | 424                                        |

## O.

|                                                       |                                           |
|-------------------------------------------------------|-------------------------------------------|
| Omaha, reticence of, as to secret societies           | 396                                       |
| Onondaga, work of J. N. B. Hewitt among               | <a href="#">XXX</a> , <a href="#">XXX</a> |
| Oqomiut Eskimo tribe, situation and subdivisions of   | 424-440                                   |
| Origin of the Adlet and the Qadlunait                 | 637                                       |
| Origin of the narwhal                                 | 625-627                                   |
| Osage Indians, traditions among                       | <a href="#">LVI</a>                       |
| Osage traditions, paper on, by J.O. Dorsey            | 373-397                                   |
| Otis, F. M., paper on Panama ornaments by, mentioned  | 46                                        |
| Owen's Valley, California, pictographic material from | <a href="#">XLII</a>                      |

## P.

|                                      |                                                                                                                            |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Padlimiut Eskimo tribe, situation of | 440-442                                                                                                                    |
| Parry, W. E., cited                  | 410, 443, 444,<br>447, 451, 458,<br>464, 474, 475,<br>478, 487, 492,<br>494, 502, 509,<br>510, 517, 523,<br>533, 544, 545, |

|                                                                                                                       |                                                                              |
|-----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
|                                                                                                                       | 547, 552, 556,<br>557, 558, 559,<br>572, 574, 603,<br>614                    |
| Penn wampum belt                                                                                                      | 233                                                                          |
| Penny, cited                                                                                                          | 425                                                                          |
| Peruvians, ancient, illustrations of ornamentation by                                                                 | 211, 212, 214,<br>228, 230, 231,<br>235, 236, 237,<br>242, 243, 248          |
| Petermanns Mitteilungen, cited                                                                                        | 409, <i>note</i>                                                             |
| Petitot, É., cited                                                                                                    | 412, 516                                                                     |
| Piedra pintal, description of, by Seemann                                                                             | 21, 22                                                                       |
| Pilingmiut Eskimo tribe, situation of                                                                                 | 444                                                                          |
| Pilling, J. C., Bibliography of the Languages of the North<br>American Indians by<br>[Pilling, J. C.], office work of | <a href="#">XXIV</a> -XXVI<br><br><a href="#">XLV</a> , <a href="#">XLVI</a> |
| Pima Indians, illustrations of ornamentation by                                                                       | 220                                                                          |
| Pinart, A. L., cited                                                                                                  | 14, 15, 20, 22                                                               |
| Piute Indians, illustrations of ornamentation by                                                                      | 198, 205                                                                     |
| Poetry and music of the Eskimo                                                                                        | 648-658                                                                      |

|                                                                 |                                                  |
|-----------------------------------------------------------------|--------------------------------------------------|
| Polychrome group of Chiriquian pottery                          | 140-147                                          |
| Polynesian ornamentation, illustrations of                      | 249, 250                                         |
| Ponka, secret society among                                     | 396                                              |
| Pottery of Chiriqui                                             | 53-186                                           |
| Powell, J.W., report of operations of Bureau of Ethnology<br>by | <a href="#">XXIII</a> -LVIII                     |
| Pueblo Indians, researches among                                | <a href="#">XXVIII</a> ,<br><a href="#">XXIX</a> |

## Q.

|                                                   |                        |
|---------------------------------------------------|------------------------|
| Qailertétang, fabulous people in Eskimo tradition | 640                    |
| Qaudjaqdjuq                                       | 628-633                |
| Qaumauangmiut Eskimo tribe, situation of          | 421, 422               |
| Qinguamiut Eskimo tribe, situation of             | 424                    |
| Quapaw Indians, linguistic work among             | <a href="#">XXXVII</a> |

## R.

|                                      |                                                                                       |
|--------------------------------------|---------------------------------------------------------------------------------------|
| Rae, John, cited                     | 411, 445, 446,<br>448, 450, 451,<br>452, 455, 459,<br>478, 485, 510,<br>597           |
| Rattles of ancient Chiriqui          | 156, 157                                                                              |
| Red line group of Chiriquian pottery | 109-111                                                                               |
| Religious ideas of the Eskimo        | 583-609                                                                               |
| Riggs, R. B., analyses by            | 49                                                                                    |
| Rink, H., cited                      | 411, 420, 580,<br>586, 587, 590,<br>591, 598, 599                                     |
| [Rink, H.], acknowledgments to       | 412                                                                                   |
| Rogan, J. P., field work of          | <a href="#">XXVI</a> , <a href="#">XXVII</a>                                          |
| Rosny, L. de, cited                  | 267, 347, 355,<br>357                                                                 |
| Ross, J., cited                      | 410, 451, 453,<br>454, 455, 456,<br>458, 469, 471,<br>478, 485, 508,<br>552, 553, 579 |
| Royce, C. C., office work of         | <a href="#">L</a> , <a href="#">LI</a>                                                |

## S.

|                                                               |                                         |
|---------------------------------------------------------------|-----------------------------------------|
| Sagdlirmiut Eskimo tribe, situation of                        | 444                                     |
| Sagdlirmiut of Southampton Island                             | 451                                     |
| San Mateo Mountains, Indians near                             | <a href="#"><u>XXXIX</u></a>            |
| Santa Barbara, Cal., pictographs at                           | <a href="#"><u>XLII</u></a>             |
| Santa Barbara Indians, linguistic work of H. W. Henshaw among | <a href="#"><u>XXXI</u></a> -<br>XXXIII |
| Saumingmiut Eskimo tribe, situation of                        | 424                                     |
| Scarified group of Chiriquian pottery                         | 87-90                                   |
| Schellhas, P., cited                                          | 345, 359, 360,<br>361, 362, 364         |
| Schultz-Sellack, K., cited                                    | 278                                     |
| Schwatka, F., cited                                           | 445, 457, 458,<br>459, 464, 465,<br>470 |
| Science and the arts among the Eskimo                         | 643-658                                 |
| Seal hunting, Eskimo method of                                | 471-501                                 |
| Sedna and the fulmar                                          | 583-587                                 |

|                                                     |                                            |
|-----------------------------------------------------|--------------------------------------------|
| Sedna feast                                         | 594                                        |
| Seemann, description of piedra pintal by            | 21, 22                                     |
| Seminole Indians, illustrations of ornamentation by | 207                                        |
| Sikosuilarmiut Eskimo tribe, situation of           | 421, 463                                   |
| Siletz Agency, linguistic work of J. O. Dorsey at   | <a href="#">XXXVI</a>                      |
| Silvestre, É., Paléographie universelle of, cited   | 267                                        |
| Simpson, J., cited                                  | 411, 597                                   |
| Simpson, T., cited                                  | 410, 458                                   |
| Singing house of Eskimo                             | 600-602                                    |
| Sinimiut Eskimo tribe, situation of                 | 451                                        |
| Sledges and boats, description of Eskimo            | 527-538                                    |
| Smith, Mrs. E. A., field work of                    | <a href="#">XXX</a> , <a href="#">XXXI</a> |
| [Smith, Mrs. E. A.], office work of                 | <a href="#">XLV</a>                        |
| [Smith, Mrs. E. A.], collections of                 | <a href="#">XLIX</a> , <a href="#">L</a>   |
| Smith Sound, Eskimo tribes of                       | 459, 460                                   |
| Snow houses, of Davis Strait Eskimo                 | 541-544                                    |
| [Snow houses], of Iglulik Eskimo tribe              | 544                                        |
| [Snow houses], of Hudson Bay Indians                | 547                                        |

|                                                                                         |                                                                          |
|-----------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Social life and customs of Eskimo                                                       | 574-578                                                                  |
| Songish Indians, burial customs and remains of                                          | <a href="#">XLII</a>                                                     |
| Spicer, J. O., acknowledgments to<br>[Spicer, J. O.], cited                             | 409<br>489, 511, 587,<br>588, 611                                        |
| Spindle whorls of Chiriqui                                                              | 149, 150                                                                 |
| Stearns, J. B., specimens in archæological collections of                               | 24, 41, 43, 45,<br>48, 49                                                |
| Stevenson, J., field work of<br>[Stevenson, J.], collection of objects of Pueblo art by | <a href="#">XXVIII</a> ,<br><a href="#">XXIX</a><br><a href="#">XLIX</a> |
| Stevenson, Mrs. J., researches among the Zuñi by                                        | <a href="#">XXIX</a>                                                     |
| Stools of ancient Chiriqui                                                              | 154-156                                                                  |
| Sturgis, A., acknowledgments to<br>[Sturgis, A.], cited                                 | 409<br>491                                                               |

## T.

|                                          |         |
|------------------------------------------|---------|
| Talirpingmiut Eskimo tribe, situation of | 424     |
| Tents of Eskimo, mode of construction of | 551-553 |
| Terra cotta group of Chiriquian pottery  | 67      |

|                                                                                               |                                                                     |
|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| Textile art in its relation to the development of form and ornament, paper by W. H. Holmes on | 189-252                                                             |
| Thing, L. H., field work of                                                                   | <a href="#"><u>XXVII</u></a> ,<br><a href="#"><u>XXVIII</u></a>     |
| Thlinkit ivory and wood carvings                                                              | <a href="#"><u>XLII</u></a>                                         |
| Thomas, C., field work of                                                                     | <a href="#"><u>XXVI</u></a> -<br>XXVIII, <a href="#"><u>XXX</u></a> |
| [Thomas, C.], office work of                                                                  | <a href="#"><u>XLVII</u></a>                                        |
| [Thomas, C.], collection from mounds by                                                       | <a href="#"><u>L</u></a>                                            |
| [Thomas, C.], abstract of paper on aids to the study of the Maya codices by                   | <a href="#"><u>LV</u></a> -LVI                                      |
| [Thomas, C.], paper on aids to the study of the Maya codices by                               | 253-371                                                             |
| Tonkawē Indians, linguistic work of A. S. Gatschet among                                      | <a href="#"><u>XXXIII</u></a> ,<br><a href="#"><u>XXXIV</u></a>     |
| Tornait and angakut                                                                           | 591-598                                                             |
| Tornit, the                                                                                   | 634-636, 640                                                        |
| Trade and intercourse between Eskimo tribes                                                   | 462-470                                                             |
| Tripod group of Chiriquian pottery                                                            | 97-107                                                              |
| Troano Manuscript, copy of, by Brasseur                                                       | 285, 286, 343                                                       |
| Tule River Indians, illustrations of ornamentation by                                         | 219                                                                 |

|                                                                                         |                                                 |
|-----------------------------------------------------------------------------------------|-------------------------------------------------|
| Tununirmiut Eskimo tribe, situation of                                                  | 442-444                                         |
| Tununirusirmiut Eskimo tribe, situation of                                              | 442-444                                         |
| Turner, L. M., cited                                                                    | 420, 462, 520,<br>565, 567, 608,<br><i>note</i> |
| Tusayan, model of the seven villages of<br>[Tusayan], collection of fictile products of | <a href="#">XLVII</a><br><a href="#">XLIX</a>   |
| Tusayan ornament, illustrations of                                                      | 247, 248                                        |

## U.

|                                                |          |
|------------------------------------------------|----------|
| Udleqdjun                                      | 636, 637 |
| Ugjulirmiut Eskimo tribe, situation of         | 458      |
| Uissuit                                        | 621      |
| [Uissuit], fabulous people in Eskimo tradition | 640      |
| Ukusiksalirmiut Eskimo tribe, situation of     | 458      |
| Ungavimiut Eskimo tribe, situation of          | 463      |

## V.

|                     |     |
|---------------------|-----|
| Vater, J. S., cited | 262 |
|---------------------|-----|

## W.

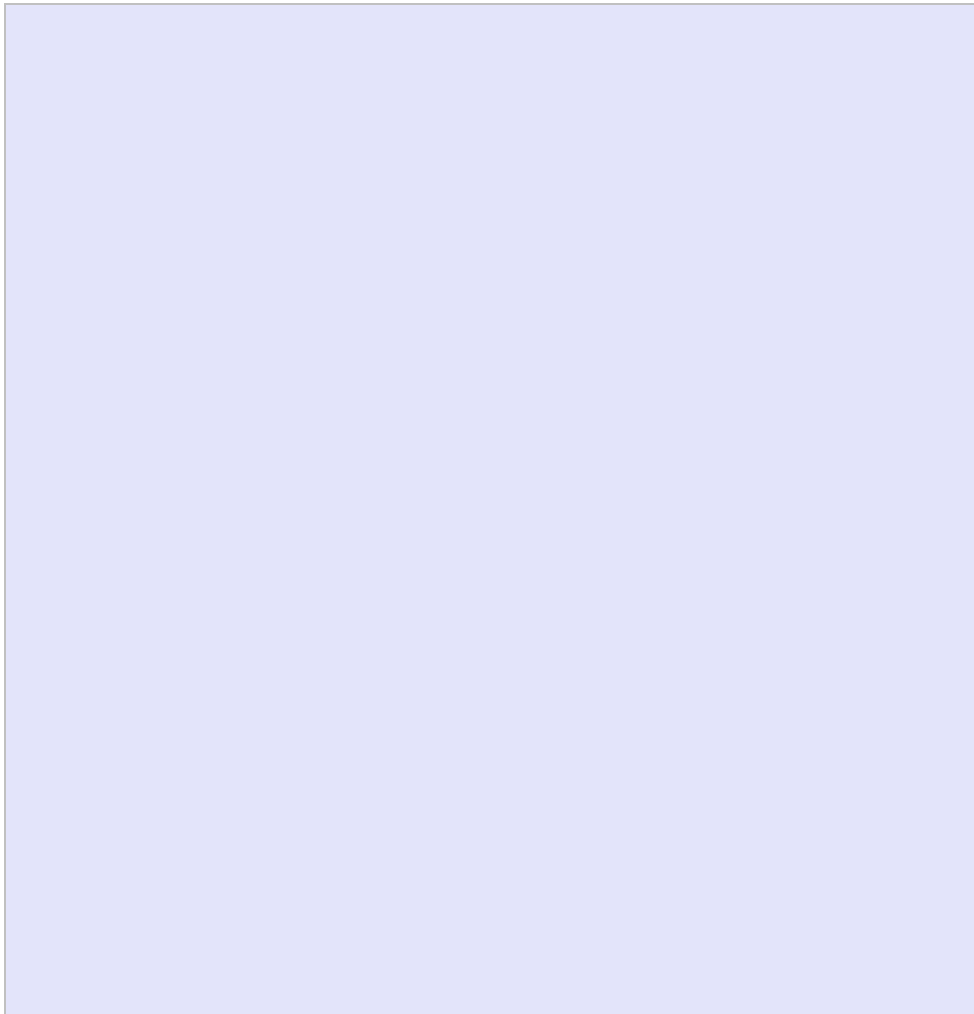
|                                                         |                       |
|---------------------------------------------------------|-----------------------|
| Walpi, model of                                         | <a href="#">XLVII</a> |
| Warmow, cited                                           | 425, 583              |
| Whistles of ancient Chiriqui                            | 164-171               |
| White, B. B., description of cemetery in New Granada by | 19                    |
| White line group of Chiriquian pottery                  | 111-113               |
| Wiener, cited                                           | 242                   |
| Wind instruments of ancient Chiriqui                    | 160-171               |
| Winnebago, sacred songs of the Iowa in                  | 396                   |

## Y.

|                                                  |                         |
|--------------------------------------------------|-------------------------|
| Yarrow, H. C., collections of                    | <a href="#">XLIX</a>    |
| [Yarrow, H. C.], field work of                   | <a href="#">XL</a> -XLI |
| [Yarrow, H. C.], office work of                  | <a href="#">L</a>       |
| Yokut Indians, illustrations of ornamentation by | 233, 234                |
| Yuchi Indians, linguistic work among             | <a href="#">XXV</a>     |

## Z.

|                                                       |                                              |
|-------------------------------------------------------|----------------------------------------------|
| Zeltner, A. de, observations on graves in Chiriqui by | 14, 18, 19, 41,<br>42                        |
| [Zeltner, A. de], cited                               | 20, 22, 27, 43,<br>45, 140                   |
| [Zeltner, A. de], description of Chiriquian vases by  | 145-147                                      |
| Zuñi, studies of Mrs. J. Stevenson among              | <a href="#">XXIX</a>                         |
| [Zuñi], culture growth of                             | <a href="#">XLVI</a> , <a href="#">XLVII</a> |
| [Zuñi], collection made at                            | <a href="#">XLIX</a>                         |
| [Zuñi], illustrations of ornamentation by             | 239                                          |



## **TRANSCRIBER'S NOTES**

1. The Bureau of Ethnology—later the Bureau of American Ethnology—published a total of 48 Annual Reports over the period 1879 to 1933. Each Report was divided into three pieces: introductory material including summaries of what everyone in the Bureau had been doing; a number of free-standing articles; and an Index covering the whole volume.
2. Each article is a separate project, and the introduction and index are combined into another project. Articles are posted separately to PG, and when a volume is finished, a whole-volume version is also posted.
3. This project contains the roman numbered introduction plus the index for the volume.
4. Each volume contains a full index for all papers in that volume. To make the Index usable, page anchors were included in a consistent format, using the physical page number.
5. The arabic numbered page anchors will be linked when the whole volume version is posted to PG. Until this happens only the roman numbered page links will be valid.
6. Silently corrected simple spelling, grammar, and typographical errors.
7. Retained anachronistic and non-standard spellings as printed.

\*\*\* END OF THE PROJECT GUTENBERG EBOOK SIXTH ANNUAL  
REPORT OF THE BUREAU OF ETHNOLOGY. (1888 N 06 / 1884-1885)

\*\*\*

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

# THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at [www.gutenberg.org/license](http://www.gutenberg.org/license).

## **Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works**

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at [www.gutenberg.org](http://www.gutenberg.org). If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website ([www.gutenberg.org](http://www.gutenberg.org)), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

## 1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

## **Section 2. Information about the Mission of Project Gutenberg**

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at [www.gutenberg.org](http://www.gutenberg.org).

### **Section 3. Information about the Project Gutenberg Literary Archive Foundation**

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at [www.gutenberg.org/contact](http://www.gutenberg.org/contact)

### **Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation**

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit [www.gutenberg.org/donate](http://www.gutenberg.org/donate).

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: [www.gutenberg.org/donate](http://www.gutenberg.org/donate).

## **Section 5. General Information About Project Gutenberg electronic works**

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:  
[www.gutenberg.org](http://www.gutenberg.org).

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.